

GAME & RIDE MODULE

Functional Design & Configuration Reference

TICVAI | MODULE 10 | SUB-MODULE 10.2

Professional Edition

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Board 1 — Game & Ride Command Center

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption

Board Purpose: Central configuration and operational control of all games, rides, and attractions that participate in TICVAI wallet, credit, entitlement, and redemption transactions.

This board establishes the **master configuration layer**. It should not duplicate the detailed reader configuration, wallet/bonus rules, pricing, or redemption configuration that will come in later boards.

The source requires wallet-based pay-as-you-go across redemption/skill games, rides and video games, together with configurable readers, game pricing, group pricing, packages and game entitlements.

Screen 1 — Game & Ride Operations Dashboard

Purpose

Provide management with one central view of the operational and configuration status of all games, rides, skill games, video games and redemption games.

UI / Configuration

Header KPIs

- Total Attractions
- Active Attractions
- Offline / Unavailable
- Active Readers
- Reader Faults
- Transactions Today
- Wallet Credits Consumed
- Redemption Credits Earned

Attraction Status Overview

- Attraction Name
- Attraction Code
- Type
- Venue / Zone
- Reader
- Current Status
- Wallet Enabled
- Current Price
- Last Transaction

Status indicators

- Active
- Inactive
- Maintenance
- Reader Offline
- Configuration Error

Actions

- Add Attraction
- Open Attraction
- Disable Attraction
- View Configuration
- View Reader
- View Transactions
- Filter by venue/zone/type/status

Acceptance Criteria

Management can identify the current operational and configuration status of games and rides from a single screen.

TICVAI Recommended Enhancement: configuration-health warnings and live device-health indicators.

Screen 2 — Game & Ride Directory

Purpose

Maintain the master catalogue of all attractions participating in the payment/redemption ecosystem.

Main Grid

Field	Description
Attraction ID	Unique system identifier
Attraction Code	Business code
Attraction Name	Display name
Type	Ride / Skill / Video / Redemption
Venue	Assigned venue
Zone	Physical area
Reader Type	Assigned reader category
Wallet	Enabled/Disabled

Field	Description
Redemption	Earn / Spend / N/A
Status	Active/Inactive/Maintenance

Filters

- Venue
- Zone
- Attraction Type
- Reader Type
- Wallet Enabled
- Redemption Enabled
- Status

Actions

- Create
- Edit
- Clone
- Activate/Deactivate
- View
- Export

Acceptance Criteria

Authorized users can search, filter and manage the complete game and ride catalogue.

Screen 3 — Attraction Profile

Purpose

Create or maintain the master record for an individual game or ride.

General Information

- Attraction ID — system generated
- Attraction Code
- Attraction Name
- Description
- Attraction Type
- Venue
- Zone / Location
- Operator
- Status

Supported Attraction Types

At minimum:

- Ride
- Skill Game
- Video Game
- Redemption Game

The source explicitly distinguishes video games, skill games and rides when defining reader configurations.

Transaction Configuration

- Wallet Payment Enabled

- Credit Payment Enabled
- Bonus Accepted
- Entitlement Accepted
- Redemption Credits Generated
- Free Play Supported

Device Association

- Assigned Reader
- Reader Profile
- Device ID
- Connection Status

Actions

Save Draft | Validate | Activate | Deactivate

Acceptance Criteria

Each game/ride has a unique master configuration that determines how it participates in TICVAI's payment and entitlement ecosystem.

Screen 4 — Attraction Type Configuration

Purpose

Configure reusable categories for games and rides so common rules do not have to be configured separately for every attraction.

Type Directory

Examples:

Ride

- Wallet payment
- Entitlement
- Free ride
- VIP pricing

Skill Game

- Wallet payment
- Retry pricing
- Redemption earning

Video Game

- Wallet payment
- Bonus usage

Redemption Game

- Wallet payment
- Redemption credit earning

Configuration Fields

- Type Name
- Type Code
- Description
- Default Reader Type
- Wallet Allowed

- Bonus Allowed
- Entitlement Allowed
- Redemption Earning
- Retry Supported
- VIP Pricing Supported
- Free Play Supported

Inheritance

Allow defaults from the attraction type to flow into individual attraction configuration.

Type Default → Attraction Override

Acceptance Criteria

Administrators can define reusable attraction categories and establish common transaction behavior.

TICVAI Recommended Enhancement: reusable attraction templates and controlled inheritance.

Screen 5 — Game & Ride Operational Configuration

Purpose

Configure whether an attraction is currently available for customer transactions.

Configuration

Operational State

- Active
- Inactive
- Maintenance

Transaction Availability

- Allow Wallet Payment
- Allow Bonus
- Allow Entitlement
- Allow Free Play
- Allow Redemption Earning

Effective Period

- Effective From
- Effective To

Operational Message

Optional customer/operator message such as:

Ride temporarily unavailable.

Governance

- Reason for status change
- Changed by
- Changed date/time

Acceptance Criteria

An authorized operator can control whether an attraction can accept customer transactions without deleting or rebuilding its configuration.

Screen 6 — Wallet & Credit Acceptance Mapping

Purpose

Define at attraction level which payment/value mechanisms are accepted.

The source specifically requires digital-wallet credits to support pay-as-you-go for redemption games, skill games, rides and video games.

Accepted Value Types

For each attraction:

Value Type	Accept	Priority
Cash Wallet Credit	✓	2
Bonus Credit	✓	1
Free Game Credit	✓	3
Game Entitlement	✓	4
Ride Entitlement	✓	4

Deduction Priority

The system must support deduction ordering.

The detailed gameplay requirement explicitly states that **bonus deduction should be done first**.

Example:

Bonus → Paid Credit → Entitlement

Detailed deduction logic will be configured under **Board 4 – Gameplay Validation & Entitlements**.

Acceptance Criteria

The attraction can identify which wallet/credit mechanisms it accepts before a gameplay transaction is authorized.

Screen 7 — Attraction / Reader Mapping

Purpose

Provide a high-level association between games/rides and their reader configurations.

Important: This screen only performs the mapping. Detailed reader properties belong to **Board 2 – Game Reader & Device Configuration**.

Mapping Table

- Attraction
- Attraction Type
- Venue / Zone
- Reader ID
- Reader Profile
- Reader Type
- Status

Reader Categories

Support at least:

- Video Game Reader
- Skill Game Reader
- Ride Reader

The source requires a minimum of three configurable reader types and specifically identifies these combinations.

Actions

- Assign Reader
- Replace Reader
- Remove Assignment
- View Reader Configuration
- Test Mapping

Validation

Prevent:

- Invalid reader profile
- Incompatible attraction type
- Duplicate assignment where prohibited
- Inactive reader assignment

Acceptance Criteria

Every applicable game/ride can be associated with the appropriate reader configuration.

Screen 8 — Game Package & Entitlement Association

Purpose

Show which packages, tickets and entitlements provide access to each game or ride.

The source requires packages containing specific games with configurable entitlement validity, plus products that can allow all games/rides, specific games/rides unlimited times, or specific games/rides a limited number of times.

Association Table

- Attraction
- Package
- Entitlement
- Access Type
- Usage Limit
- Validity
- Status

Access Types

- Pay Per Play
- Included in Package
- Free Play
- Unlimited Play
- Limited Play

Example

Adventure Package

- Roller Coaster → Unlimited
- VR Game → 2 Plays
- Basketball Skill Game → 3 Plays
- Crane Game → Pay Per Play

Actions

- View Package

- Associate Package
- Remove Association
- View Entitlement
- Check Impact

Acceptance Criteria

Operators can determine which products/packages grant access to each attraction and under what usage model.

Detailed package creation and entitlement logic will remain in **Board 4**, avoiding duplicate configuration.

Screen 9 — Configuration Health & Validation

Purpose

Prevent incomplete or conflicting game/ride configurations from becoming operational.

Validation Checks

System checks:

Attraction

- Attraction type assigned
- Venue/zone assigned
- Active configuration

Reader

- Reader assigned
- Reader active
- Compatible reader type

Payment

- Accepted value configured
- Wallet/payment rule available

Pricing

- Valid price available

Entitlement

- Valid package/entitlement mapping

Redemption

- Redemption earning configuration available where applicable

Health Status



Ready



Warning



Configuration Error

Example Error

Basketball Skill Game

No active reader assigned.

Roller Coaster

Wallet enabled but no valid price configuration found.

Actions

- Fix Issue
- Open Configuration
- Validate Again
- View Dependencies

Acceptance Criteria

The system identifies missing or conflicting configuration before an attraction is activated.

TICVAI Recommended Enhancement: AI-assisted configuration diagnosis with suggested fixes.

Screen 10 — Attraction Audit, Dependencies & Governed Actions

Purpose

Provide traceability and controlled management of attraction configuration changes.

Audit Timeline

Capture:

- Attraction created
- Type changed
- Reader assigned/replaced
- Wallet acceptance changed
- Package association changed
- Operational status changed
- Activation/deactivation
- User
- Date/time
- Previous value
- New value

Dependency View

Show relationships:

Attraction

- Reader
- Wallet/Credit Rules
- Pricing
- Packages
- Entitlements
- Redemption
- Inventory, where applicable

Governed Actions

- Activate
- Deactivate
- Put into Maintenance
- Clone
- Archive
- Restore

Impact Warning

Example:

Deactivate “VR Racing”?

3 active packages and 2 entitlement rules currently reference this attraction.

Acceptance Criteria

All significant attraction configuration changes are auditable, and the system warns authorized users about affected dependencies before critical changes.

TICVAI Recommended Enhancement: graphical dependency map and impact analysis.

Board 1 — Final Screen Mapping

#	Screen	Primary Role
1	Game & Ride Operations Dashboard	Central overview
2	Game & Ride Directory	Master catalogue
3	Attraction Profile	Individual game/ride setup
4	Attraction Type Configuration	Reusable game/ride categories
5	Game & Ride Operational Configuration	Availability/status
6	Wallet & Credit Acceptance Mapping	Accepted value mechanisms
7	Attraction / Reader Mapping	Device association
8	Game Package & Entitlement Association	Product/game relationships
9	Configuration Health & Validation	Readiness/error control
10	Attraction Audit, Dependencies & Governed Actions	Governance & traceability

Requirements covered by Board 1

Board 1 establishes the foundation for **10.2.1, 10.2.6, 10.2.7, 10.2.8, 10.2.11 and 10.2.12**, while only referencing the detailed functionality that will be configured on subsequent boards.

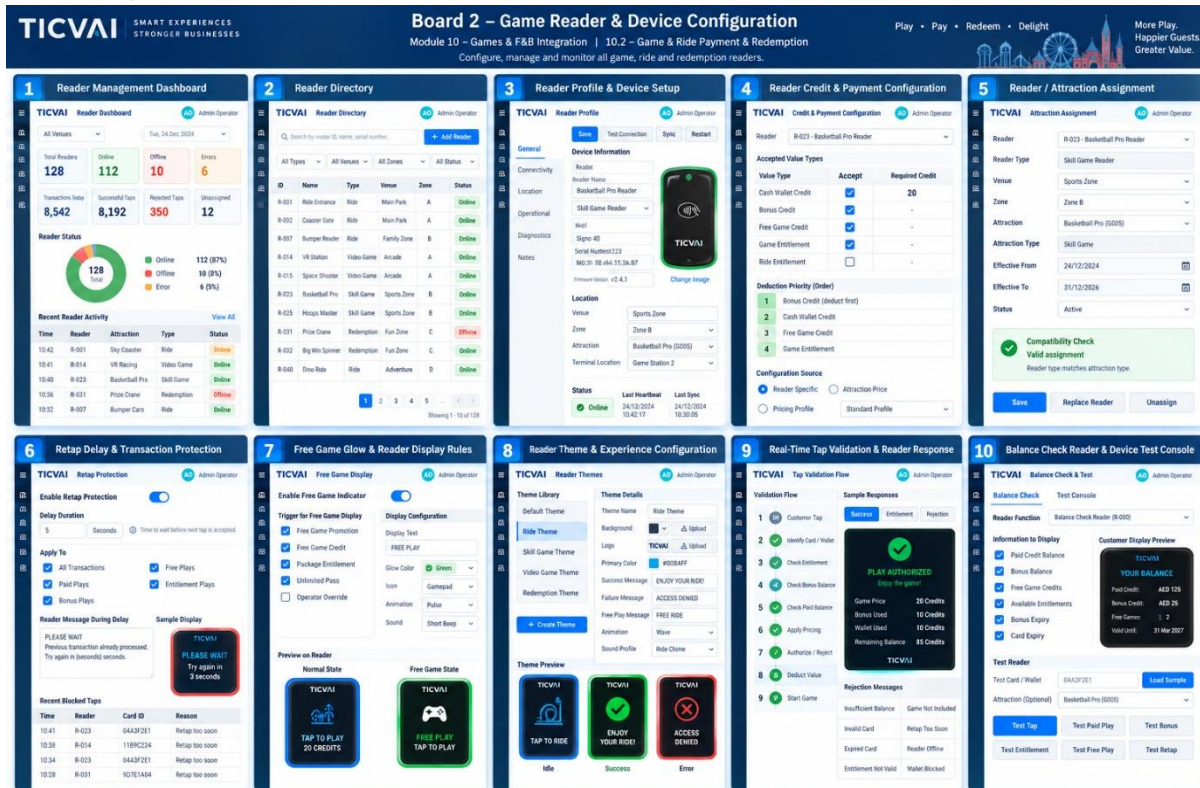
The important rule for the **Board 1 visual design** is now fixed: **the image must contain exactly these 10 screens, with these same screen names and functions.** I will not substitute generic dashboard screens or introduce unrelated screens when designing it.

Board 2 — Game Reader & Device Configuration

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption

Board Purpose: Configure, manage, deploy, and monitor the physical readers used across games, rides, skill games, video games, redemption games, and balance-check stations.



Board 2 is especially important because the source specifically requires **multiple reader types with configurable properties**, including the credit type accepted, amount required, and attraction type linked to each reader. It also requires real-time tap validation, free-game color indication, retap delay, custom reader themes, and dedicated balance-check readers.

The detailed Board 2 design must therefore contain **exactly the following 10 screens**.

Screen 1 — Reader Management Dashboard

Purpose

Provide a centralized operational overview of all game and ride readers deployed across venues.

KPI Cards

- Total Readers
- Online Readers
- Offline Readers
- Readers with Errors
- Unassigned Readers
- Transactions Today
- Successful Taps
- Rejected Taps

Reader Status Overview

Reader	Type	Attraction	Zone	Status	Last Tap
R-001	Ride Reader	Sky Coaster	Zone A	Online	10:42
R-014	Video Reader	VR Racing	Arcade	Online	10:41
R-023	Skill Reader	Basketball Pro	Sports	Online	10:40
R-031	Redemption	Prize Crane	Fun Zone	Offline	10:15

Filters

- Venue
- Zone
- Reader Type
- Attraction Type
- Status
- Assigned / Unassigned

Actions

[Add Reader](#) | [Configure](#) | [Test](#) | [Disable](#) | [View Transactions](#)

Acceptance Criteria

Authorized users can see the operational and configuration status of all readers from one central screen.

Screen 2 — Reader Directory

Purpose

Maintain the master inventory of physical readers connected to TICVAI.

The source requires the ability to define **a minimum of three reader types**.

Directory Columns

- Reader ID
- Reader Name
- Reader Type
- Serial Number
- Venue
- Zone
- Assigned Attraction
- Firmware Version
- Connection Status
- Configuration Status
- Last Communication

Reader Types

At minimum:

- Ride Reader
- Skill Game Reader
- Video Game Reader

The architecture should additionally accommodate:

- Redemption Game Reader
- Balance Check Reader

The last two are required by other source functions, including redemption processing and customer balance checking.

Actions

- Add Reader
- Edit
- Clone Configuration
- Assign
- Unassign
- Disable
- Search
- Export

Acceptance Criteria

Administrators can maintain a centralized registry of all readers and their current assignments.

Screen 3 — Reader Profile & Device Setup

Purpose

Configure the technical and operational identity of an individual reader.

Device Information

- Reader ID
- Reader Name
- Reader Type
- Manufacturer
- Model
- Serial Number
- MAC / Device Identifier
- Firmware Version

Location

- Client
- Venue
- Zone
- Attraction
- Terminal Location

Connectivity

- Connection Type
- IP Address
- Port
- Online / Offline
- Last Heartbeat
- Last Synchronization

Operational Settings

- Enabled
- Transaction Enabled
- Offline Mode Allowed
- Logging Enabled

Actions

[Save](#) | [Test Connection](#) | [Synchronize](#) | [Restart](#) | [Disable](#)

Acceptance Criteria

Every reader has a uniquely identifiable technical profile and can be associated with its physical deployment location.

TICVAI Recommended Enhancement: remote device diagnostics and firmware visibility.

Screen 4 — Reader Credit & Payment Configuration

Purpose

Define exactly what type of wallet value a reader accepts and how much is required to start the game or ride.

This directly implements requirement 10.2.6, which requires configurable:

- Credit type accepted
- Credit amount required
- Attraction type linked to the reader.

Configuration

Reader: R-023 – Basketball Skill Game

Accepted Value Types

Value Type	Accept
Paid Wallet Credit	✓
Bonus Credit	✓
Free Game Credit	✓
Game Entitlement	✓
Ride Entitlement	—

Required Credit

Standard Play Price: AED 20 / 20 Credits

Deduction Priority

1. Bonus Credit
2. Paid Credit
3. Free Credit / Entitlement as applicable

The source specifically requires **bonus deduction first** during gameplay validation.

Configuration Source

- Reader-specific
- Attraction price
- Pricing profile

Acceptance Criteria

Each reader can determine what customer value it accepts and the required amount for gameplay authorization.

Screen 5 — Reader / Attraction Assignment

Purpose

Assign a physical reader to the appropriate game, ride, or attraction.

Assignment Form

- Reader
- Reader Type
- Venue
- Zone
- Attraction
- Attraction Type
- Effective From
- Effective To
- Status

Compatibility Validation

Example:

Reader: R-023

Reader Type: Skill Game

Attraction: Basketball Pro

Attraction Type: Skill Game

Compatibility: ✓ Valid

Invalid example:

Ride Reader cannot be assigned to this Video Game under the selected configuration profile.

Actions

Assign | Replace Reader | Unassign | Validate | Save

Acceptance Criteria

Readers can be mapped to attractions, and incompatible reader/attraction combinations can be identified before activation.

Screen 6 — Retap Delay & Transaction Protection

Purpose

Prevent accidental repeated deductions when a customer taps the card multiple times within a short period.

The requirement explicitly states that a configurable delay—example **5 seconds**—must exist between the first tap and subsequent taps.

Configuration

Retap Protection: ON/OFF

Delay Duration

- Seconds
- Example: 5 seconds

Apply To

- All Transactions
- Paid Plays
- Bonus Plays

- Free Plays
- Entitlement Plays

Reader Response During Delay

Example:

PLEASE WAIT

Previous transaction already processed.

Try again in 3 seconds.

Logging

- First Tap
- Blocked Retap
- Reader
- Card/Wallet Reference
- Timestamp

Acceptance Criteria

A second tap received within the configured delay does not create an unintended second charge.

Screen 7 — Free Game Glow & Reader Display Rules

Purpose

Configure how the reader visually communicates that a game or ride is available free of charge.

The source specifically requires free games to display a **different color on the reader**, allowing the customer to recognize that the game is free.

Configuration

Free Game Indicator: Enabled

Trigger

- Free Game Promotion
- Free Game Credit
- Package Entitlement
- Unlimited Pass
- Operator Override

Display Configuration

- Free Game Glow
- Display Text
- Icon
- Animation
- Sound

Example preview:

FREE PLAY

TAP TO PLAY

Default/Free State Preview

The screen design should visually show:

NORMAL STATE → FREE GAME STATE

Acceptance Criteria

The reader can visibly differentiate a free-play game from a normally charged game.

Exact physical LED capabilities will depend on the selected reader hardware/API.

Screen 8 — Reader Theme & Experience Configuration

Purpose

Configure different visual experiences for reader categories.

The source requires **custom themes controlled from BOS**, with skill games and rides able to have different reader themes.

Theme Library

Examples:

Ride Theme

Skill Game Theme

Video Game Theme

Redemption Theme

Theme Configuration

- Theme Name
- Background
- Logo
- Reader Screen Layout
- Icons
- Display Messages
- Animation
- Sound Profile
- Success State
- Failure State
- Free Play State

Assignment

Theme → Reader Type → Readers

Example:

Skill Game Theme

→ 42 Skill Game Readers

Preview

The configuration screen must show a visual reader preview.

Actions

Create Theme | Edit | Preview | Publish | Assign

Acceptance Criteria

BOS administrators can centrally configure and deploy different reader themes by game/ride category.

Screen 9 — Real-Time Tap Validation & Reader Response

Purpose

Define and visualize what occurs when a customer taps their card/device before gameplay.

This screen directly addresses requirement 10.2.13: the system must validate the customer's **credit/card balance, bonus and entitlements in real time before the game starts.**

Validation Flow

Customer Tap

↓

Identify Card / Wallet

↓

Check Entitlement

↓

Check Bonus

↓

Check Paid Balance

↓

Apply Pricing

↓

Authorize / Reject

↓

Deduct Value

↓

Start Game

Successful Response

✓ PLAY AUTHORIZED

Game Price: AED 20

Bonus Used: AED 10

Wallet Used: AED 10

Remaining Balance: AED 85

Entitlement Response

✓ INCLUDED IN YOUR PASS

Remaining Plays: 2

Rejection Examples

- Insufficient Balance
- Invalid Card
- Expired Card
- Entitlement Not Valid
- Game Not Included
- Retap Too Soon
- Reader Offline
- Wallet Blocked

Acceptance Criteria

The reader receives a clear authorization/rejection result before allowing gameplay, and successful transactions update the applicable balance/entitlement.

Screen 10 — Balance Check Reader & Device Test Console

Purpose

Configure dedicated balance-check readers and provide administrators with a test environment for deployed reader configurations.

Requirement 10.2.20 specifically requires **a reader through which the customer can check their balance.**

Balance Reader Configuration

Reader Function: Balance Check

Customer Display

Upon tap, display:

- Paid Credit Balance
- Bonus Balance
- Free Game Credits
- Available Entitlements
- Bonus Expiry
- Card Expiry

Example:

YOUR BALANCE

Paid Credit: **AED 125**

Bonus Credit: **AED 25**

Free Games: **2**

Valid Until: **31 Mar 2027**

Administrative Test Console

Select:

- Reader
- Test Card / Wallet
- Attraction
- Simulated Balance
- Entitlement

Test Actions

- Test Tap
- Test Paid Play
- Test Bonus
- Test Entitlement
- Test Free Play
- Test Retap
- Test Rejection

Result

Show:

- Validation result
- Deduction source

- Price
- Remaining balance
- Reader response
- Response time

Acceptance Criteria

A dedicated reader can display customer balances without initiating gameplay, and authorized administrators can test reader configuration before deployment.

TICVAI Recommended Enhancement: configuration simulation/test console.

Board 2 — Exact Design Mapping

The visual Board 2 must contain these **exact 10 screens in this exact order**:

#	Screen
1	Reader Management Dashboard
2	Reader Directory
3	Reader Profile & Device Setup
4	Reader Credit & Payment Configuration
5	Reader / Attraction Assignment
6	Retap Delay & Transaction Protection
7	Free Game Glow & Reader Display Rules
8	Reader Theme & Experience Configuration
9	Real-Time Tap Validation & Reader Response
10	Balance Check Reader & Device Test Console

Requirement coverage

Board 2 primarily covers **10.2.6, 10.2.7, 10.2.13, 10.2.14, 10.2.18, 10.2.19 and 10.2.20**.

For the **Board 2 design**, I will use this mapping literally. For example, **Screen 6 will be the Retap Delay configuration, Screen 7 will visibly show the free-game reader/glow states, Screen 8 will show actual reader-theme previews, and Screen 9 will visualize the complete tap-validation sequence** rather than replacing them with generic administration screens.

Board 3 — Wallet, Credits & Bonus Management

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption

Board Purpose: Configure the complete wallet value model used for games and rides, including paid credits, bonus credits, free-game/ride credits, top-ups, usage restrictions, validity, refunds, and manual bonus allocation.



Board 3 directly addresses the requirements for digital-wallet pay-as-you-go, free-game credits, real-time balance usage, and especially **Card Top-Up + Bonus**, where a top-up such as AED 100 can provide AED 100 paid value + AED 25 bonus while restricting the bonus to games/rides and preventing refund of the bonus.

The visual Board 3 must later contain **exactly these 10 screens**.

Screen 1 — Wallet & Credit Management Dashboard

Purpose

Provide a central management view of wallet balances, credit issuance, bonus utilization, free-game credits and top-up activity across the operation.

KPI Cards

- Active Wallets
- Total Paid Credit Balance
- Total Bonus Balance
- Free Game/Ride Credits
- Top-Ups Today
- Bonus Issued Today
- Bonus Used Today
- Expiring Bonus

Wallet Activity

Show:

- Top-Ups

- Bonus Issuance
- Bonus Consumption
- Free Credits
- Game/Ride Spending
- Expired Credits

Credit Distribution

Break down wallet value into:

Paid Credit | Bonus Credit | Free Game Credit | Entitlements

Filters

- Venue
- Date
- Wallet Type
- Credit Type
- Customer
- Status

Actions

Wallet Search | Configure Credits | Configure Top-Up | Add Bonus | View Transactions

Acceptance Criteria

Authorized users can understand current wallet and credit activity without navigating through individual wallets.

Screen 2 — Wallet & Credit Type Configuration

Purpose

Define the different value buckets that can exist inside a guest's digital wallet.

The source distinguishes normal wallet value, bonus value, free-game credits and game/ride entitlements.

Credit Type Library

Credit Type	Monetary	Refundable	Restricted	Status
Paid Credit	Yes	Configurable	No	Active
Bonus Credit	Yes	No	Yes	Active
Free Game Credit	No	No	Games/Rides	Active
Redemption Credit	No	No	Redemption	Active

Configuration Fields

- Credit Type Name
- Code
- Description
- Monetary / Non-Monetary
- Usage Scope
- Refundable
- Transferable
- Validity Required
- Priority
- Active / Inactive

Usage Scope

Examples:

- Games
- Rides
- Skill Games
- Video Games
- Redemption
- F&B
- Retail

Acceptance Criteria

Administrators can define and maintain distinct wallet value types with controlled usage behavior.

Redemption-credit accumulation and prize redemption will be configured in detail under **Board 6**, rather than duplicated here.

Screen 3 — Wallet Account & Balance View

Purpose

Allow authorized operators to inspect all value held in an individual guest wallet.

The source requires stored credits to be viewable through operator and self-service kiosks.

Wallet Header

- Wallet ID
- Customer
- Card / RFID ID
- Wallet Status
- Created Date
- Last Activity
- Last Recharge

Balance Cards

Paid Credit

AED 125

Bonus Credit

AED 25

Free Games

2

Redemption Credits

1,450

Entitlements

- All Games Pass
- Ride Pass
- Specific Game Entitlements
- Remaining Plays

Expiry Information

For each balance:

- Expiry Date
- Days Remaining
- Status

Actions

- View Transactions
- Add Credit — permission controlled
- Add Bonus — permission controlled
- View Entitlements
- Block Wallet

Acceptance Criteria

The operator can clearly distinguish paid, bonus, free, redemption and entitlement balances belonging to the same wallet.

Screen 4 — Top-Up Configuration

Purpose

Configure the rules governing customer wallet recharges.

Top-Up Rule

- Rule Name
- Minimum Top-Up
- Maximum Top-Up
- Fixed Amounts / Open Amount
- Currency
- Applicable Venue
- Applicable Channel
- Effective From / To
- Status

Channels

- POS
- Self-Service Kiosk
- B2C
- Mobile
- Operator
- Other integrated channel

Example

Top-Up	Wallet Value
AED 50	AED 50
AED 100	AED 100
AED 200	AED 200

Bonus is configured separately in Screen 5.

Actions

[Create Rule](#) | [Edit](#) | [Activate](#) | [Suspend](#) | [Clone](#)

Acceptance Criteria

Administrators can define permitted wallet top-up values and their applicable channels/effective periods.

Screen 5 — Top-Up Bonus Rule Configuration

Purpose

Configure promotional bonus value automatically awarded when a customer tops up.

The requirement explicitly gives the example:

Top-Up AED 100 → AED 100 Paid Credit + AED 25 Bonus.

Bonus Rule

Rule Name: AED 100 + AED 25 Bonus

Trigger

- Minimum Recharge: AED 100
- Maximum Recharge: optional
- Exact Amount / Threshold
- Channel
- Venue
- Effective Period

Bonus Calculation

Choose:

- Fixed Bonus
- Percentage Bonus

Example:

Top-Up: AED 100

Paid Balance Added: AED 100

Bonus Added: AED 25

Total Wallet Increase: AED 125

Tier Configuration

Recharge	Bonus
AED 100	AED 25
AED 200	AED 60
AED 500	AED 175

Actions

[Add Tier](#) | [Preview](#) | [Validate](#) | [Save](#) | [Activate](#)

Acceptance Criteria

The system automatically calculates and allocates the correct bonus based on the configured top-up rule.

Screen 6 — Bonus Usage Restrictions

Purpose

Control where promotional bonus value can and cannot be spent.

This is a critical requirement because the source states that the bonus can be used for **amusement park games/rides but not for F&B or Retail**, and that this must be configurable.

Allowed Categories

- ☒ Games
- ☒ Rides
- ☒ Skill Games
- ☒ Video Games

Restricted Categories

- ☒ F&B
- ☒ Retail
- ☒ Prize Redemption

Additional Controls

- Venue
- Zone
- Attraction
- Attraction Type
- Reader Type

Rule Example

BONUS25 can be consumed only at Games & Rides readers.

Reader Enforcement

When a customer attempts to use bonus at an unauthorized location:

BONUS NOT ACCEPTED

System should preserve the bonus and evaluate another valid value source where applicable.

Acceptance Criteria

Bonus value cannot be consumed outside its configured business scope.

Screen 7 — Bonus Validity & Expiry Configuration

Purpose

Define how long promotional bonus credit remains valid.

The source specifically requires the ability to set a **validity period for the bonus**.

Validity Methods

- Fixed Expiry Date
- X Days from Issue
- X Days from Top-Up
- Promotional Period

Example

Bonus Validity: 30 days from issuance

Issued: 01 Sep 2026

Expires: 30 Sep 2026

Expiry Behavior

- Stop usage immediately at expiry
- Mark value as expired
- Maintain audit history

Monitoring

- Bonus Expiring Today
- Within 7 Days
- Within 30 Days
- Expired

Acceptance Criteria

Every bonus allocation follows the configured validity rule and becomes unusable after expiry.

Screen 8 — Free Game & Ride Credit Management

Purpose

Configure and allocate free gameplay credits to customer wallets.

The requirement explicitly requires the system to add **credits for free games and rides to a guest's digital wallet**, usable across applicable attractions.

Free Credit Configuration

- Credit Name
- Game / Ride
- Attraction Type
- Venue
- Number of Free Plays
- Unlimited / Limited
- Valid From
- Valid Until

Example

Birthday Free Play

Applicable:

- VR Racing
- Basketball Pro
- Bumper Cars

Usage:

3 Free Plays

Allocation

- Individual Customer
- Campaign/Promotion
- Product/Package
- Customer Service
- Bulk Allocation

Actions

[Create](#) | [Allocate](#) | [Revoke Unused](#) | [View Usage](#)

Acceptance Criteria

Authorized users can issue free-game/ride credits and restrict them to applicable attractions and validity periods.

Screen 9 — Refund, Adjustment & Manual Bonus Control

Purpose

Enforce refund restrictions and provide governed manual wallet adjustments.

The requirement states:

- remaining balance is not refundable under the specified bonus/top-up model;
- bonus cannot be refunded;
- additional bonus may be provided to the customer on demand.

Adjustment Types

- Manual Bonus
- Promotional Credit
- Free Game
- Correction
- Administrative Adjustment

Manual Bonus Form

- Wallet
- Bonus Amount
- Reason
- Usage Rule
- Validity
- Requested By
- Approval Status

Refund Control

Paid Credit

→ Apply configured refund policy.

Bonus Credit

→ **Non-Refundable**

Warning Example

Bonus credit cannot be refunded.

Governance

High-value manual adjustments can require:

- Supervisor approval
- Reason code
- Comment
- Audit record

Acceptance Criteria

Bonus cannot be refunded, while authorized staff can provide additional bonus under controlled permissions and audit rules.

TICVAI Recommended Enhancement: configurable approval thresholds for manual bonus adjustments.

Screen 10 — Wallet Credit Transaction Ledger & Audit

Purpose

Provide complete traceability for every change to wallet value.

Transaction Ledger

Time	Type	Source	Debit	Credit	Balance
10:00	Top-Up	POS-03	—	100	100
10:00	Bonus	BONUS25	—	25	25 Bonus
10:20	Game Play	VR Racing	20	—	80
10:35	Bonus Play	Basketball	10	—	15 Bonus

Transaction Types

- Top-Up
- Bonus Allocation
- Bonus Consumption
- Game/Ride Purchase
- Free Play
- Expiry
- Manual Adjustment
- Reversal
- Redemption Credit Movement

Details

Each transaction should retain:

- Transaction ID
- Wallet
- Card
- Reader
- Attraction
- Operator
- Channel
- Value Type
- Before Balance
- Amount
- After Balance
- Date/Time
- Reason

Search

- Wallet
- Card
- Customer
- Transaction ID
- Reader
- Attraction
- Date
- Transaction Type

Acceptance Criteria

Every credit issuance, deduction, expiry or adjustment can be traced to its source and resulting balance.

Board 3 — Exact Screen Mapping

The visual design for **Board 3** must use these exact ten screens:

#	Screen
1	Wallet & Credit Management Dashboard
2	Wallet & Credit Type Configuration
3	Wallet Account & Balance View
4	Top-Up Configuration
5	Top-Up Bonus Rule Configuration
6	Bonus Usage Restrictions
7	Bonus Validity & Expiry Configuration
8	Free Game & Ride Credit Management
9	Refund, Adjustment & Manual Bonus Control
10	Wallet Credit Transaction Ledger & Audit

Requirement coverage

Board 3 primarily covers **10.2.1, 10.2.3, 10.2.5, 10.2.7, 10.2.13 and 10.2.22**, with detailed redemption-credit processing deliberately left for the dedicated redemption board.

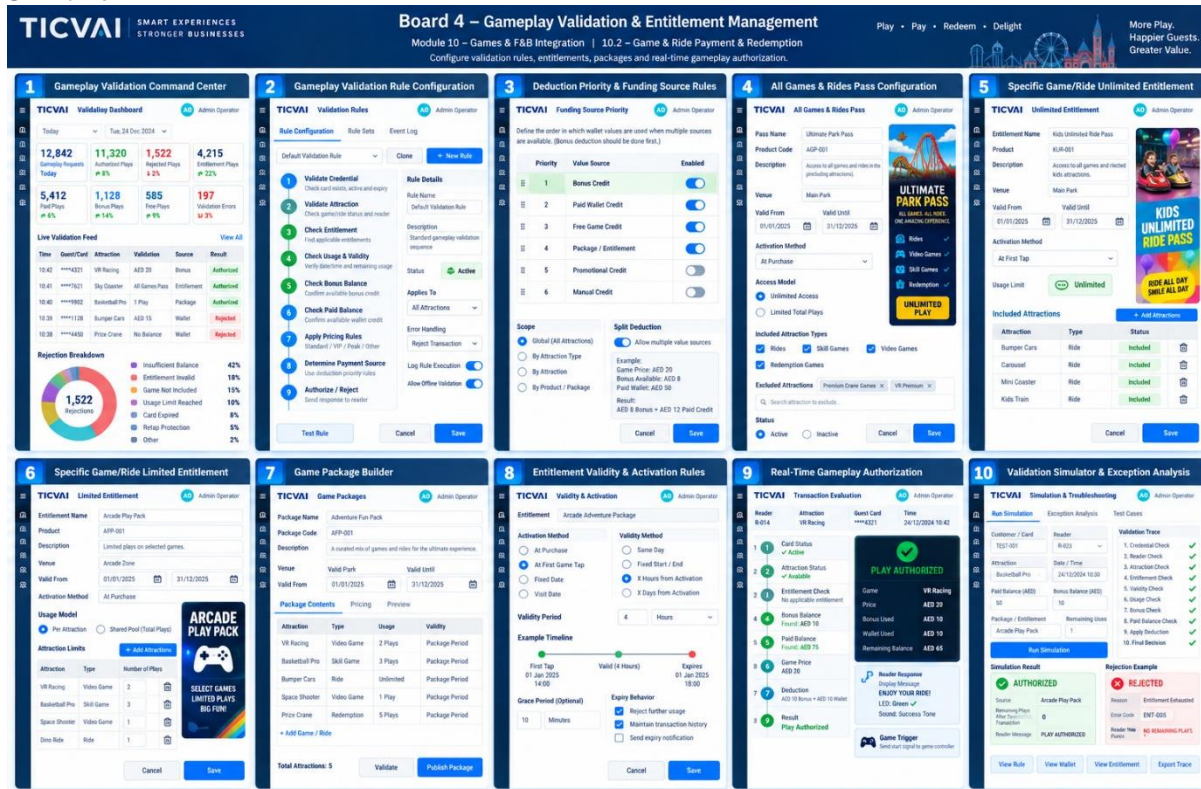
For the next **Board 3 design**, the key requirement is that the visual screens **5, 6 and 7 clearly show three different concepts—how the bonus is earned, where it can be spent, and when it expires**. This separation is important because these are three independent business configurations, even though all three relate to the same bonus wallet.

Board 4 — Gameplay Validation & Entitlement Management

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption

Board Purpose: Configure and control the decision engine that determines whether a guest is entitled to play a specific game or ride when they tap their RFID card/digital credential. The board covers real-time validation, deduction priority, all-games passes, specific-game access, limited/unlimited usage, packages, entitlement validity, and the resulting gameplay authorization.



This is one of the **core transaction boards**. The source specifically requires real-time validation of **credit/card balance, bonus and entitlements before the game starts**, as well as packages, validity, unlimited/limited free-play products, and bonus-first deduction.

As agreed, the eventual visual design must contain **exactly these 10 screens**.

Screen 1 — Gameplay Validation Command Center

Purpose

Provide an operational overview of gameplay authorization across all games and rides.

KPI Cards

- Gameplay Requests Today
- Authorized Plays
- Rejected Plays
- Entitlement Plays
- Paid Plays
- Bonus Plays
- Free Plays
- Validation Errors

Live Validation Feed

Time	Guest/Card	Attraction	Validation	Source	Result
10:42	****4321	VR Racing	AED 20	Bonus	Authorized
10:41	****7621	Sky Coaster	All Games Pass	Entitlement	Authorized
10:40	****9902	Basketball Pro	1 Play	Package	Authorized
10:39	****1128	Bumper Cars	AED 15	Wallet	Rejected

Rejection Breakdown

- Insufficient Balance
- Entitlement Invalid
- Game Not Included
- Entitlement Expired
- Usage Limit Reached
- Card Expired
- Retap Protection

Filters

- Venue
- Zone
- Attraction
- Validation Type
- Result
- Date/Time

Acceptance Criteria

Operators can monitor gameplay validation results and immediately identify abnormal rejection patterns.

Screen 2 — Gameplay Validation Rule Configuration

Purpose

Configure the checks TICVAI performs before allowing a game or ride to start.

The source requires validation **before game start** against balance, bonus and applicable entitlements.

Validation Checks

Credential

- Card/credential exists
- Card is active
- Card has not expired

Attraction

- Game/ride active
- Reader valid
- Attraction included

Entitlement

- Entitlement exists
- Entitlement active
- Within validity

- Remaining usage available

Value

- Bonus available
- Paid balance available
- Required price available

Rule Sequence

Tap

- Validate Credential
- Validate Attraction
- Find Entitlement
- Check Usage/Validity
- Check Bonus
- Check Paid Balance
- Determine Payment Source
- Authorize / Reject

Actions

[Create Rule](#) | [Change Sequence](#) | [Test Rule](#) | [Save](#) | [Activate](#)

Acceptance Criteria

Validation logic can be configured and executed consistently before gameplay authorization.

Screen 3 — Deduction Priority & Funding Source Rules

Purpose

Determine which wallet/value source TICVAI consumes when more than one valid funding source is available.

Requirement 10.2.7 explicitly states:

“Bonus deduction should be done first.”

Priority Configuration

Example:

Priority	Value Source	Enabled
1	Bonus Credit	✓
2	Paid Wallet Credit	✓
3	Free Game Credit	✓
4	Package/Entitlement	✓

Controls

- Drag to reorder
- Enable/disable source
- Apply by attraction
- Apply by attraction type
- Apply by product/package

Split Deduction

Configurable:

Allow multiple value sources for one transaction: ON/OFF

Example:

Game Price: AED 20

Bonus Available: AED 8

Paid Wallet: AED 50

Result:

AED 8 Bonus + AED 12 Paid Credit

Acceptance Criteria

The engine applies the configured funding priority and uses bonus first where the business rule requires it.

Screen 4 — All Games & Rides Pass Configuration

Purpose

Configure a ticket/product that permits the guest to play **all games/rides in the amusement park**.

This directly represents requirement 10.2.8.

Pass Setup

- Pass Name
- Product Code
- Venue
- Valid From
- Valid Until
- Activation Method
- Included Attraction Types
- Excluded Attractions
- Status

Access Model

All Games / Rides

Options:

- Unlimited
- Limited Total Plays

Example

Ultimate Park Pass

Valid: 09 Sep 2026

Includes:

- Rides ✓
- Video Games ✓
- Skill Games ✓

Excluded:

- Premium Crane Games

Reader Behavior

When tapped:

✓ INCLUDED IN YOUR PASS

TAP ACCEPTED

Acceptance Criteria

A valid all-games ticket allows access to all applicable attractions without requiring individual paid-wallet deductions.

Screen 5 — Specific Game/Ride Unlimited Entitlement

Purpose

Configure products that provide **unlimited free usage of selected games or rides**.

This directly implements requirement 10.2.11.

Entitlement Configuration

- Entitlement Name
- Product
- Venue
- Included Games
- Included Rides
- Validity
- Activation
- Unlimited Usage = ON
- Status

Attraction Selection

Example:

Kids Unlimited Ride Pass

- ✓ Bumper Cars
- ✓ Carousel
- ✓ Mini Coaster
- ✓ Kids Train

Usage

Usage Limit: UNLIMITED

Reader Result

FREE PLAY

Included in your pass.

Acceptance Criteria

The guest can repeatedly use selected games/rides while the entitlement remains valid.

Screen 6 — Specific Game/Ride Limited Entitlement

Purpose

Configure products providing a defined number of free plays on specific attractions.

This directly implements requirement 10.2.12.

Configuration

- Entitlement Name

- Product
- Included Attractions
- Number of Plays
- Usage Scope
- Validity
- Status

Usage Models

Per Attraction

VR Racing → 2 plays

Basketball Pro → 3 plays

Bumper Cars → 1 play

or

Shared Pool

5 plays across selected attractions.

Real-Time Usage

Basketball Pro

Allowed: 3

Used: 2

Remaining: **1**

Reader Response

✓ PLAY AUTHORIZED

Remaining Plays: **0**

Acceptance Criteria

Each successful gameplay transaction reduces the applicable usage allowance and prevents additional plays once the limit reaches zero.

Screen 7 — Game Package Builder

Purpose

Create a package containing a specific collection of games/rides.

Requirement 10.2.7 explicitly requires:

- Create a package with specific games.
- Define validity for the game entitlement.

Package Header

- Package Name
- Package Code
- Venue
- Description
- Effective Dates
- Status

Package Contents

Attraction	Type	Usage
VR Racing	Video Game	2
Basketball Pro	Skill Game	3
Bumper Cars	Ride	Unlimited
Space Shooter	Video Game	1

Configuration

For each attraction:

- Include / Exclude
- Limited / Unlimited
- Number of Plays
- Validity
- Consumption Group

Actions

[Add Game](#) | [Add Ride](#) | [Remove](#) | [Preview](#) | [Validate](#) | [Publish](#)

Acceptance Criteria

Administrators can create reusable packages containing selected games/rides with different usage allowances.

Screen 8 — Entitlement Validity & Activation Rules

Purpose

Control when a gameplay entitlement becomes valid and when it expires.

The source specifically requires the ability to **define validity for game entitlements**.

Activation Methods

- At Purchase
- At First Game Tap
- Fixed Date
- Visit Date

Validity Methods

- Same Day
- Fixed Start/End
- X Hours from Activation
- X Days from Activation

Example

Arcade Adventure Package

Activation: First Tap

Validity: 4 Hours

First Tap: 14:00

Expiry: 18:00

Expiry Behavior

- Reject further usage

- Preserve transaction history
- Mark entitlement Expired

Grace Period

Optional configurable grace period.

Acceptance Criteria

The validation engine determines entitlement validity using its configured activation and expiration rules.

Screen 9 — Real-Time Gameplay Authorization

Purpose

Provide the operational transaction view showing exactly how TICVAI evaluates a customer tap.

Requirement 10.2.13 requires the system to validate the balance/card, bonus and entitlement in real time and deduct the applicable amount before gameplay.

Transaction Example

Reader: R-014

Game: VR Racing

Guest Card: ****4321

Validation Steps

1. Card Status

✓ Active

2. Attraction Status

✓ Available

3. Entitlement Check

✗ No applicable entitlement

4. Bonus Balance

✓ AED 10

5. Paid Balance

✓ AED 75

6. Game Price

AED 20

7. Deduction

Bonus: AED 10

Paid Credit: AED 10

Result

✓ **PLAY AUTHORIZED**

Remaining Paid Balance: **AED 65**

Alternative Entitlement Result

✓ **PLAY AUTHORIZED**

Package: Arcade Adventure

Remaining Plays: **1**

Acceptance Criteria

The authorization result identifies the validation checks, funding source, amount consumed, remaining value and final decision.

Screen 10 — Validation Simulator & Exception Analysis

Purpose

Allow administrators to test entitlement and payment configurations before deployment and investigate rejected gameplay transactions.

Simulation Inputs

- Customer/Card
- Reader
- Attraction
- Date/Time
- Package
- Entitlement
- Paid Balance
- Bonus Balance

Run Simulation

Example:

Card: Test-001

Game: VR Racing

Entitlement: Arcade Package

Remaining Uses: 1

Decision Trace

1. Credential ✓
2. Reader ✓
3. Attraction ✓
4. Package ✓
5. Validity ✓
6. Usage Remaining ✓
7. Authorization ✓

Result

AUTHORIZED

Source: Arcade Package

Remaining Plays After Transaction: **0**

Rejected Example

REJECTED — ENTITLEMENT EXHAUSTED

Exception Investigation

For actual rejected transactions show:

- Transaction ID
- Reader
- Card
- Rule evaluated
- Failed condition
- Rejection code
- Timestamp

Actions

[Simulate](#) | [View Rule](#) | [View Wallet](#) | [View Entitlement](#) | [Export Trace](#)

Acceptance Criteria

Administrators can test configuration before activation and trace exactly why an actual gameplay transaction was authorized or rejected.

TICVAI Recommended Enhancement: AI-assisted exception diagnosis identifying likely configuration errors and recommending corrective actions.

Board 4 — Exact Screen Mapping

The Board 4 visual must contain **exactly these ten screens and in this order**:

#	Screen
1	Gameplay Validation Command Center
2	Gameplay Validation Rule Configuration
3	Deduction Priority & Funding Source Rules
4	All Games & Rides Pass Configuration
5	Specific Game/Ride Unlimited Entitlement
6	Specific Game/Ride Limited Entitlement
7	Game Package Builder
8	Entitlement Validity & Activation Rules
9	Real-Time Gameplay Authorization
10	Validation Simulator & Exception Analysis

Critical logic that must appear in the design

I want to lock these details now so that **Board 4's picture follows the SOW exactly**:

Screen 3 must visually show **Bonus as Priority 1**. This is an explicit source requirement, not merely a TICVAI recommendation.

Screens 4, 5 and 6 must remain visibly different: Screen 4 is an **all-games/rides pass**, Screen 5 is **selected attractions with unlimited usage**, and Screen 6 is **selected attractions with limited usage**. These correspond to separate requirements in the supplied matrix.

Screen 7 must actually display several games/rides being added to a package with individual usage allowances—not a generic package form.

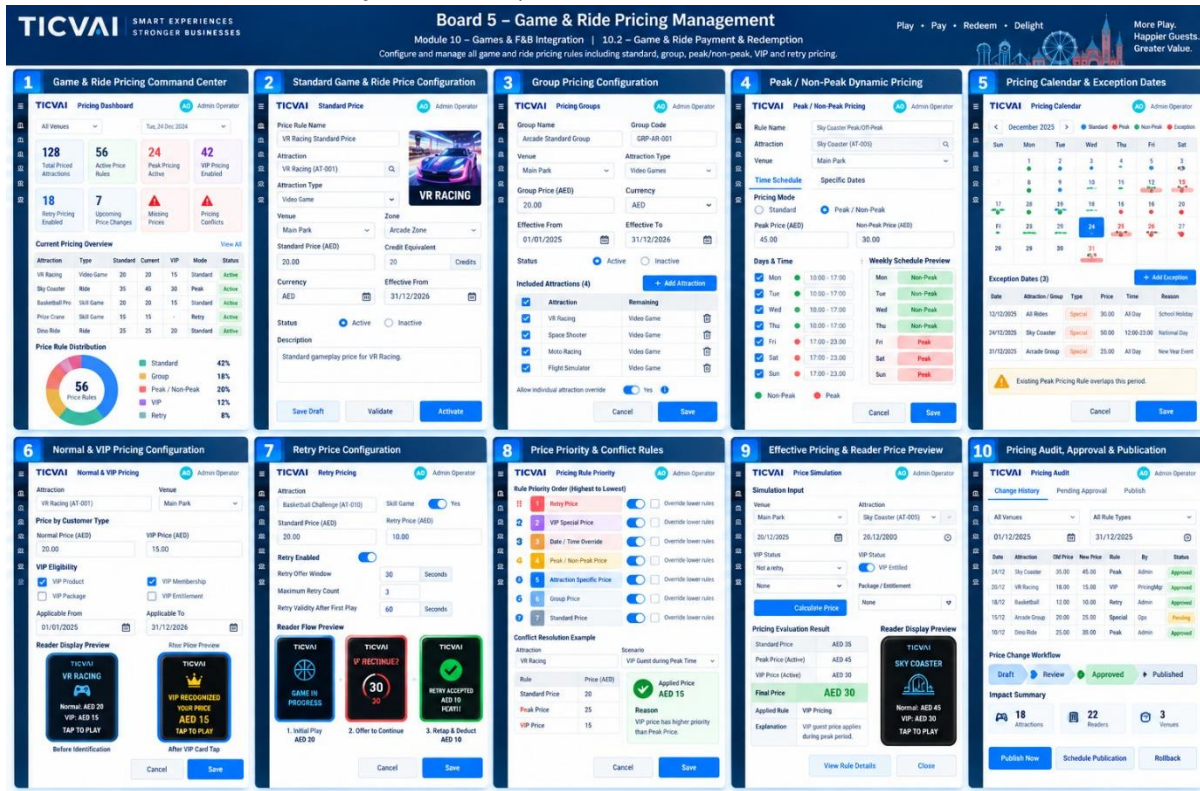
Screen 9 must show the actual transaction decision: **card → entitlement → bonus → paid balance → deduction → authorization**, including remaining balance. That reflects the real-time gameplay-validation requirement.

This board therefore becomes the **central rules engine for deciding “Can this guest play this game right now, and what should TICVAI consume?”**

Board 5 — Game & Ride Pricing Management

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption



Board Purpose: Configure and govern all pricing rules applied to games and rides, including standard price, group price, peak/non-peak pricing, retry pricing, Normal/VIP pricing, effective dates, priority, conflict handling, and pre-publication simulation.

This board is fully backend-driven. The physical reader does **not** calculate the business price itself. TICVAI determines the applicable price based on the configured rules and sends the result/authorization to the reader.

The source explicitly requires:

- price per game;
- group pricing;
- dynamic peak/non-peak pricing by date/time;
- reduced retry price for skill games;
- Normal and VIP prices on the same reader.

The visual Board 5 must contain **exactly the following 10 screens**.

Screen 1 — Game & Ride Pricing Command Center

Purpose

Provide a central view of all game and ride pricing configurations, active pricing rules, upcoming changes, and pricing issues.

KPI Cards

- Total Priced Attractions
- Active Price Rules
- Peak Pricing Active
- VIP Pricing Enabled
- Retry Pricing Enabled

- Upcoming Price Changes
- Missing Prices
- Pricing Conflicts

Pricing Overview

Attraction	Type	Standard	Current Price	VIP	Pricing Mode	Status
VR Racing	Video Game	AED 20	AED 20	AED 15	Standard	Active
Sky Coaster	Ride	AED 35	AED 45	AED 30	Peak	Active
Basketball Pro	Skill Game	AED 20	AED 20	AED 15	Standard	Active
Prize Crane	Skill Game	AED 15	AED 15	—	Retry Enabled	Active

Filters

- Venue
- Zone
- Attraction
- Attraction Type
- Pricing Type
- VIP Enabled
- Effective Date
- Status

Actions

[Create Price](#) | [Edit Price](#) | [View Calendar](#) | [Validate](#) | [Publish](#)

Acceptance Criteria

Administrators can immediately identify current game/ride prices, active special pricing, upcoming changes, and missing/conflicting configurations.

Screen 2 — Standard Game & Ride Price Configuration

Purpose

Define the normal base price charged to play a specific game or ride.

The source requires that **prices be defined for each game**.

Configuration

- Price Rule Name
- Attraction
- Attraction Type
- Venue
- Zone
- Standard Price
- Currency
- Wallet/Credit Equivalent
- Effective From
- Effective To
- Status

Example

VR Racing

Standard Price: **AED 20**

Credit Equivalent: **20 Credits**

Reader Relationship

The reader receives the current applicable gameplay price from TICVAI configuration/runtime services.

Actions

Save Draft | Validate | Activate | Clone

Acceptance Criteria

Every payable game/ride can have a standard base price with controlled effective dates.

Screen 3 — Group Pricing Configuration

Purpose

Allow multiple games or rides to share a common pricing rule instead of configuring each attraction individually.

The source specifically states:

Assign prices for the Group.

Group Setup

- Pricing Group Name
- Group Code
- Venue
- Attraction Type
- Price
- Currency
- Effective Period
- Status

Included Attractions

Example:

Arcade Standard Group — AED 20

- VR Racing
- Space Shooter
- Moto Racing
- Flight Simulator

Configuration Options

- Add Attraction
- Remove Attraction
- Apply price to all
- Individual override allowed: Yes/No

Inheritance Model

Group Price → Attraction

If an attraction has an approved override:

Attraction Override > Group Price

Acceptance Criteria

A price can be assigned centrally to a group of games/rides and inherited by all linked attractions.

Screen 4 — Peak / Non-Peak Dynamic Pricing

Purpose

Configure different game/ride prices based on date and time.

The source explicitly requires peak/non-peak pricing for games/rides based on **specific date/time**.

Pricing Modes

- Standard
- Peak
- Non-Peak

Rule Configuration

- Rule Name
- Attraction / Group
- Venue
- Days of Week
- Specific Date
- Start Time
- End Time
- Peak Price
- Non-Peak Price
- Effective From / To

Example

Sky Coaster

Standard: AED 35

Peak

Friday–Sunday

17:00–23:00

AED 45

Non-Peak

Monday–Thursday

10:00–17:00

AED 30

Calendar View

Visually show:

Mon | Tue | Wed | Thu | Fri | Sat | Sun

with peak/non-peak periods.

Acceptance Criteria

The backend automatically determines the applicable price using configured date/time rules.

Screen 5 — Pricing Calendar & Exception Dates

Purpose

Provide a visual calendar for understanding and overriding time-based pricing.

Calendar Layers

- Standard Pricing
- Peak
- Non-Peak
- Special Day
- Temporary Override

Exception Examples

- Public Holiday
- School Holiday
- Special Event
- Promotional Day
- Venue-Specific Date

Exception Configuration

- Date
- Attraction / Group
- Pricing Type
- Price
- Start Time
- End Time
- Priority
- Reason

Example

National Day

Sky Coaster

12:00–23:00

Special Price: AED 50

Conflict Warning

Existing Peak Pricing Rule overlaps this period.

Acceptance Criteria

Authorized users can see price schedules visually and configure date-specific exceptions without changing the underlying standard rule.

TICVAI Recommended Enhancement: exception calendar, because the source requires date/time pricing but does not explicitly prescribe the UI method.

Screen 6 — Normal & VIP Pricing Configuration

Purpose

Configure different prices for Normal and VIP guests for the same game/ride and reader.

The source explicitly requires **one reader to show both Normal Price and VIP Price for a specific game**, with VIP eligibility obtained through purchase of a VIP product.

Configuration

Attraction: VR Racing

Customer Type	Price
Normal	AED 20
VIP	AED 15

VIP Eligibility

- VIP Product
- VIP Membership
- VIP Package
- VIP Entitlement

VIP Rule

- Product / entitlement
- Valid From
- Valid To
- Applicable Games/Rides
- Status

Reader Display Logic

Before identification:

Normal AED 20 | VIP AED 15

After VIP card tap:

VIP RECOGNIZED

YOUR PRICE: AED 15

Acceptance Criteria

The same game and reader can support Normal and VIP prices, and TICVAI automatically selects the appropriate price after identifying the guest's VIP entitlement.

Screen 7 — Retry Price Configuration

Purpose

Configure a discounted repeat-play price for applicable skill games.

Requirement 10.2.16 describes a skill game where, before the game ends, the guest is offered:

“Do you want to continue?”

A subsequent RFID tap should deduct a lower amount than the original price.

Configuration

- Attraction
- Skill Game = Yes
- Standard Price
- Retry Price
- Retry Enabled
- Retry Offer Window
- Maximum Retry Count
- Retry Validity After First Play

Example

Basketball Challenge

Original Price: AED 20

Retry Price: **AED 10**

Retry Window: **30 seconds**

Runtime Flow

Initial Play — AED 20

↓

Game approaching completion

↓

Reader displays:

CONTINUE?

RETRY AED 10

↓

Guest taps card

↓

TICVAI validates balance and retry eligibility

↓

AED 10 deducted

↓

Game continues/restarts.

Acceptance Criteria

An eligible retry transaction is recognized separately from a new gameplay transaction and charged at the configured reduced price.

Screen 8 — Price Priority & Conflict Rules

Purpose

Define which pricing rule wins when several valid prices apply simultaneously.

This is required to make the source pricing models operational because Standard, Group, Peak, VIP and Retry prices can overlap.

Example Priority

1. Retry Price
2. VIP Special Price
3. Date/Time Override
4. Peak / Non-Peak Price
5. Attraction-Specific Price
6. Group Price
7. Standard Price

Configuration

For each pricing rule:

- Priority

- Stackable: Yes/No
- Override Lower Priority
- Customer Type
- Effective Period

Conflict Detection

Example:

VR Racing

VIP Price: AED 15

Peak Price: AED 25

System determines according to configured rule:

Applied Price: AED 15

Reason:

VIP price has higher priority than Peak Price.

Actions

[Reorder](#) | [Validate](#) | [Test](#) | [Save](#)

Acceptance Criteria

When multiple pricing rules apply, TICVAI consistently calculates one deterministic final price.

TICVAI Recommended Enhancement: formal priority/conflict engine. The matrix requires several overlapping price types but does not state the resolution hierarchy.

Screen 9 — Effective Pricing & Reader Price Preview

Purpose

Allow an administrator to preview what price will actually be presented/applied for a selected game before publishing configuration.

Simulation Inputs

- Venue
- Attraction
- Date
- Time
- Customer Type
- VIP Status
- Retry Status
- Package/Entitlement

Example

Attraction: Sky Coaster

Date: Saturday

Time: 20:00

Customer: VIP

Rule Evaluation

Standard Price: AED 35

Peak Rule: AED 45

VIP Rule: AED 30

Final Result

AED 30

Applied Rule: VIP Pricing

Reader Preview

Reader mockup should show:

SKY COASTER

Normal: AED 45

VIP: AED 30

TAP TO PLAY

Acceptance Criteria

Administrators can confirm the final calculated price and expected reader display before publishing configuration.

Screen 10 — Pricing Audit, Approval & Publication

Purpose

Govern pricing changes and maintain a full historical record.

Price Change History

- Rule
- Attraction
- Previous Price
- New Price
- Changed By
- Date/Time
- Reason
- Approval Status

Example

Attraction	Old	New	Rule	User
Sky Coaster	35	45	Peak	Admin
VR Racing VIP	18	15	VIP	Pricing Manager
Basketball Retry	12	10	Retry	Admin

Workflow

Draft → Review → Approved → Published

Publication

Allow:

- Publish Now
- Schedule Publication
- Roll Back
- Deactivate Rule

Impact Summary

Before publishing:

This pricing update affects:

18 attractions
22 readers
3 venues

Acceptance Criteria

Every pricing change is traceable and only approved/effective pricing is delivered to runtime services/readers.

TICVAI Recommended Enhancement: governed approval and rollback workflow.

Board 5 — Exact Screen Mapping

The visual Board 5 must contain these **exact 10 screens in this order**:

#	Screen
1	Game & Ride Pricing Command Center
2	Standard Game & Ride Price Configuration
3	Group Pricing Configuration
4	Peak / Non-Peak Dynamic Pricing
5	Pricing Calendar & Exception Dates
6	Normal & VIP Pricing Configuration
7	Retry Price Configuration
8	Price Priority & Conflict Rules
9	Effective Pricing & Reader Price Preview
10	Pricing Audit, Approval & Publication

Critical design rules for Board 5

Screen 4 must visibly show date/time-based Peak and Non-Peak configuration. It cannot be a generic price form because this is an explicit requirement.

Screen 6 must show Normal and VIP prices together for the same attraction/reader. This is important because the requirement specifically asks for one reader supporting both prices.

Screen 7 must show a real retry use case, such as:

Original AED 20 → “Continue?” → Retap → Retry AED 10

This directly reflects the source's retry transaction requirement.

Screen 9 should visually connect the backend pricing decision with the physical reader. TICVAI calculates the effective price; the reader simply displays/executes the result.

So from the architecture perspective, Board 5 answers one key question:

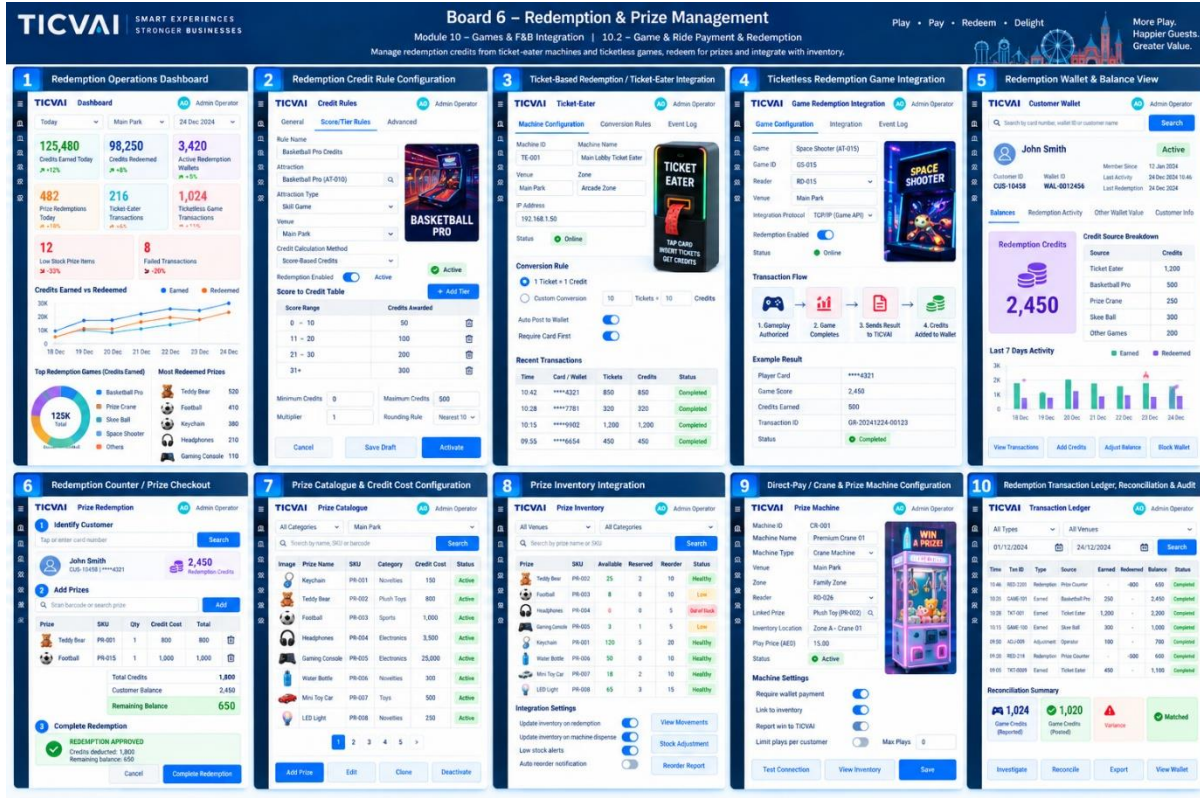
“When this customer taps this reader at this game, at this date/time and under these conditions, what price should TICVAI charge?”

Board 6 — Redemption & Prize Management

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption

Board Purpose: Manage the complete redemption ecosystem, from earning redemption credits through ticket-based or ticketless games, to storing those credits in the guest wallet, redeeming them for prizes, and synchronizing prize consumption with inventory.



This board is fully aligned with your clarified architecture:

Game / Ticket-Eater / Reader → TICVAI Backend → Wallet Redemption Credits → Redemption Counter → Prize Inventory

The source explicitly requires both **ticket-based redemption via ticket-eater machines** and **ticketless redemption from games**, with credits loaded into the guest wallet. It also requires those credits to be viewable and redeemable for prizes, with prize items integrated with inventory.

The Board 6 design must contain **exactly these 10 screens**.

Screen 1 — Redemption Operations Dashboard

Purpose

Provide a central operational view of redemption activity across venues.

KPI Cards

- Redemption Credits Earned Today
- Redemption Credits Redeemed
- Active Redemption Wallets
- Prize Redemptions Today
- Ticket-Eater Transactions
- Ticketless Redemption Transactions
- Low-Stock Prize Items
- Failed Redemption Transactions

Redemption Activity Overview

Time	Customer	Source	Credits Earned	Credits Spent	Status
10:35	Wallet 00124	Basketball Game	250	—	Completed
10:40	Wallet 00721	Ticket Eater	1,200	—	Completed
10:46	Wallet 00318	Prize Counter	—	800	Completed

Charts

- Credits Earned vs Redeemed
- Top Redemption Games
- Most Redeemed Prizes
- Credit Liability / Outstanding Balance

Filters

- Venue
- Zone
- Source Type
- Game
- Prize
- Date
- Transaction Status

Actions

[View Redemption Transactions](#) | [View Prize Inventory](#) | [View Wallet](#) | [Configure Redemption](#)

Acceptance Criteria

Operators can understand redemption-credit earning and prize-redemption activity from one central screen.

Screen 2 — Redemption Credit Rule Configuration

Purpose

Configure how games generate redemption credits.

Configuration

- Rule Name
- Game / Game Group
- Venue
- Redemption Enabled
- Credit Calculation Method
- Effective Dates
- Status

Credit Calculation Methods

Support:

- Fixed Credits Per Play
- Score-Based Credits
- Tier-Based Credits
- External Game Value
- Machine-Reported Value

Example

Basketball Skill Game

Score 0–10 → 50 credits

Score 11–20 → 100 credits

Score 21–30 → 200 credits

Score 31+ → 300 credits

Controls

- Minimum Credits
- Maximum Credits
- Multiplier
- Rounding Rule
- Bonus Redemption Credits
- Promotional Multiplier

Acceptance Criteria

TICVAI can determine how many redemption credits should be awarded for an eligible gameplay transaction.

TICVAI Recommended Enhancement: score/tier-based redemption rules because the source requires ticketless loading from games but does not define the calculation method.

Screen 3 — Ticket-Based Redemption / Ticket-Eater Integration

Purpose

Configure the integration with ticket-eater machines that convert physical redemption tickets into digital wallet credits.

The source specifically requires:

Integration with Ticket-eater machines to add credits for physical inserted redemption tickets.

Machine Configuration

- Ticket-Eater ID
- Machine Name
- Venue
- Zone
- IP / Device ID
- Status
- Reader/Scanner
- Conversion Rule

Conversion

Example:

1 Physical Ticket = 1 Redemption Credit

or configurable:

10 Tickets = 10 Credits

Transaction Flow

Customer identifies wallet/card

↓

Physical tickets inserted

↓

Ticket-Eater counts tickets

↓

Machine sends count to TICVAI

↓

TICVAI validates transaction

↓

Credits added to wallet

↓

Balance displayed**Transaction Result**Tickets Inserted: **850**Credits Added: **850**New Redemption Balance: **2,450****Exception Handling**

- Ticket jam
- Invalid ticket
- Duplicate transaction
- Machine offline
- Wallet identification failed

Acceptance Criteria

Physical tickets accepted by an integrated ticket-eater are securely converted into digital redemption credits and posted to the correct guest wallet.

Screen 4 — Ticketless Redemption Game Integration**Purpose**

Configure games that automatically send redemption-credit results to TICVAI after gameplay.

The source requires integration with redemption games to **digitally load credits after the guest completes a game**.

Game Integration

- Game
- Game ID
- Reader
- Device/Game Controller
- Venue
- Integration Protocol
- Redemption Enabled
- Credit Source
- Status

Transaction Flow**Guest taps card**

↓

Gameplay authorized

↓

Game starts

↓

Game completes

↓

Game calculates win/score

↓

Game sends redemption result

↓

TICVAI validates

↓

Credits added to wallet**Example**

Game: Basketball Pro

Score: 26

Credits Earned: **200****Validation**

Check:

- Valid authorized gameplay session
- Correct game/session ID
- Duplicate event protection
- Valid wallet
- Valid credit rule

Acceptance Criteria

Ticketless games can send redemption results to TICVAI and automatically update the guest's wallet.

Screen 5 — Redemption Wallet & Balance View**Purpose**

Allow operators and systems to view the redemption-credit balance stored within a customer wallet.

The source requires redemption credits stored in the wallet to be viewable from **operator kiosks and self-service kiosks**.

Wallet Header

- Customer
- Wallet ID
- Card / RFID
- Wallet Status

Balance

Redemption Credits: 2,450

Credit Sources

Source	Credits
Ticket Eater	1,200
Basketball Pro	500
Prize Crane	250
Other Games	500

Activity

- Earned
- Redeemed
- Adjusted
- Expired, if applicable

Display Channels

- Redemption Counter
- Operator Kiosk
- Self-Service Kiosk
- Balance Reader
- Customer App, where applicable

Acceptance Criteria

The customer's current redemption-credit balance can be retrieved consistently across supported channels.

Screen 6 — Redemption Counter / Prize Checkout

Purpose

Provide the operator interface used when a customer exchanges redemption credits for prizes.

The source explicitly requires redemption-credit usage at a **redemption counter for prizes**.

Transaction Flow

Scan / Tap Customer Card

↓

Display:

Redemption Balance: 2,450

Prize Basket

Prize	Qty	Credit Cost
Teddy Bear	1	800
Football	1	1,000

Total: 1,800 Credits

Remaining: 650

Actions

- Scan Prize Barcode
- Search Prize
- Change Quantity
- Remove Item
- Complete Redemption
- Cancel Transaction

Validation

Before completing:

- Wallet active
- Sufficient redemption credits
- Prize in stock
- Prize active
- Correct venue
- Redemption rule valid

Result

REDEMPTION APPROVED

Credits Deducted: 1,800

Remaining Balance: 650

Acceptance Criteria

The redemption counter can deduct the correct credits and issue only valid, available prizes.

Screen 7 — Prize Catalogue & Credit Cost Configuration

Purpose

Configure the prize catalogue and how many redemption credits each prize requires.

Prize Master Data

- Prize ID
- SKU
- Barcode
- Prize Name
- Category
- Description
- Image
- Venue
- Inventory Location
- Credit Cost
- Status

Example

Prize	Credit Cost
Keychain	150
Plush Toy	800
Football	1,000

Prize	Credit Cost
Headphones	3,500
Gaming Console	25,000

Controls

- Standard Credit Cost
- Promotional Credit Cost
- Effective Dates
- Venue-specific Cost
- Minimum Customer Level, optional

Actions

Add Prize | Edit | Clone | Activate | Deactivate

Acceptance Criteria

Every redeemable prize has a controlled redemption-credit cost and active/inactive status.

Screen 8 — Prize Inventory Integration

Purpose

Connect prize redemption directly with TICVAI inventory so redemption transactions reduce stock in real time.

The source explicitly requires prizes at the redemption counter, skill games and direct-pay machines such as cranes to be **integrated with the inventory system**.

Inventory View

Prize	Available	Reserved	Reorder Level	Status
Plush Toy	25	2	10	Healthy
Football	8	0	10	Low
Headphones	0	0	5	Out of Stock

Integration Rules

When redemption completed:

Prize Redeemed

↓

Inventory –1

↓

Stock Updated

↓

Low Stock Check

↓

Alert / Reorder Workflow

Support

- Redemption Counter

- Skill Game Prize
- Direct Pay Machine
- Crane Machine

Validation

Do not complete redemption if:

- Stock = 0
- Item disabled
- Inventory integration unavailable, depending on configured policy

Acceptance Criteria

Prize stock is automatically adjusted when a prize is redeemed or dispensed.

Screen 9 — Direct-Pay / Crane & Prize Machine Configuration

Purpose

Configure prize machines such as cranes or other direct-pay devices that may consume wallet value and dispense inventory items.

The matrix specifically references **skill games and direct pay machines (e.g., cranes)** as part of the inventory integration requirement.

Machine Profile

- Machine ID
- Machine Name
- Type
- Venue
- Zone
- Reader
- Linked Prize/SKU
- Inventory Location
- Play Price
- Status

Machine Types

- Crane
- Capsule Machine
- Direct Prize Machine
- Skill Prize Machine

Example

Premium Crane 01

Play Price: AED 15

Prize Inventory: Plush Toy

Current Stock: 18

Runtime Flow

Customer Tap

- TICVAI authorizes play
- Machine starts
- Result received

→ Prize won?

Yes

→ Inventory –1

Acceptance Criteria

Direct-pay prize machines can be associated with both wallet payment and prize inventory.

Screen 10 — Redemption Transaction Ledger, Reconciliation & Audit

Purpose

Provide complete traceability of redemption credits earned, redeemed, adjusted and associated inventory movements.

Transaction Ledger

Time	Transaction	Source	Earned	Redeemed	Balance
10:00	TKT-001	Ticket Eater	1,200	—	1,200
10:30	GAME-101	Basketball	200	—	1,400
11:15	RED-220	Prize Counter	—	800	600

Transaction Details

- Transaction ID
- Wallet
- Card
- Game/Machine
- Venue
- Credits Before
- Credits Earned/Spent
- Credits After
- Prize, if applicable
- Inventory Transaction ID
- Device
- Operator
- Timestamp
- Status

Reconciliation

Compare:

- Game-reported credits
- TICVAI-posted credits
- Ticket-eater counts
- Wallet balance movements
- Prize transactions
- Inventory deductions

Exceptions

- Duplicate credit award
- Missing wallet posting
- Inventory mismatch
- Machine count mismatch
- Failed redemption

- Reversed transaction

Actions

[Investigate](#) | [Reconcile](#) | [Export](#) | [View Wallet](#) | [View Inventory Transaction](#)

Acceptance Criteria

Every redemption-credit and prize transaction can be traced and reconciled against the originating device and resulting wallet/inventory movement.

TICVAI Recommended Enhancement: automated reconciliation and exception detection.

Board 6 — Exact Screen Mapping

The Board 6 visual design must contain **exactly these ten screens in this order**:

#	Screen
1	Redemption Operations Dashboard
2	Redemption Credit Rule Configuration
3	Ticket-Based Redemption / Ticket-Eater Integration
4	Ticketless Redemption Game Integration
5	Redemption Wallet & Balance View
6	Redemption Counter / Prize Checkout
7	Prize Catalogue & Credit Cost Configuration
8	Prize Inventory Integration
9	Direct-Pay / Crane & Prize Machine Configuration
10	Redemption Transaction Ledger, Reconciliation & Audit

Critical architecture for Board 6

This board should make the distinction between **three different devices/processes** very clear:

Ticket-Eater

→ Converts physical tickets into digital redemption credits.

Ticketless Redemption Game

→ Sends the result directly to TICVAI after gameplay and TICVAI adds credits.

Redemption Counter / Prize Machine

→ Consumes redemption credits and reduces inventory.

That matches the matrix requirement very closely.

And importantly, the Chinese reader remains an edge device where relevant. **TICVAI remains responsible for the wallet, redemption-credit logic, prize rules, inventory integration, reconciliation and audit.**

For the Board 6 design, I would make **Screens 3 and 4 especially visual**, showing the two separate flows:

Physical Tickets → Ticket Eater → TICVAI → Wallet

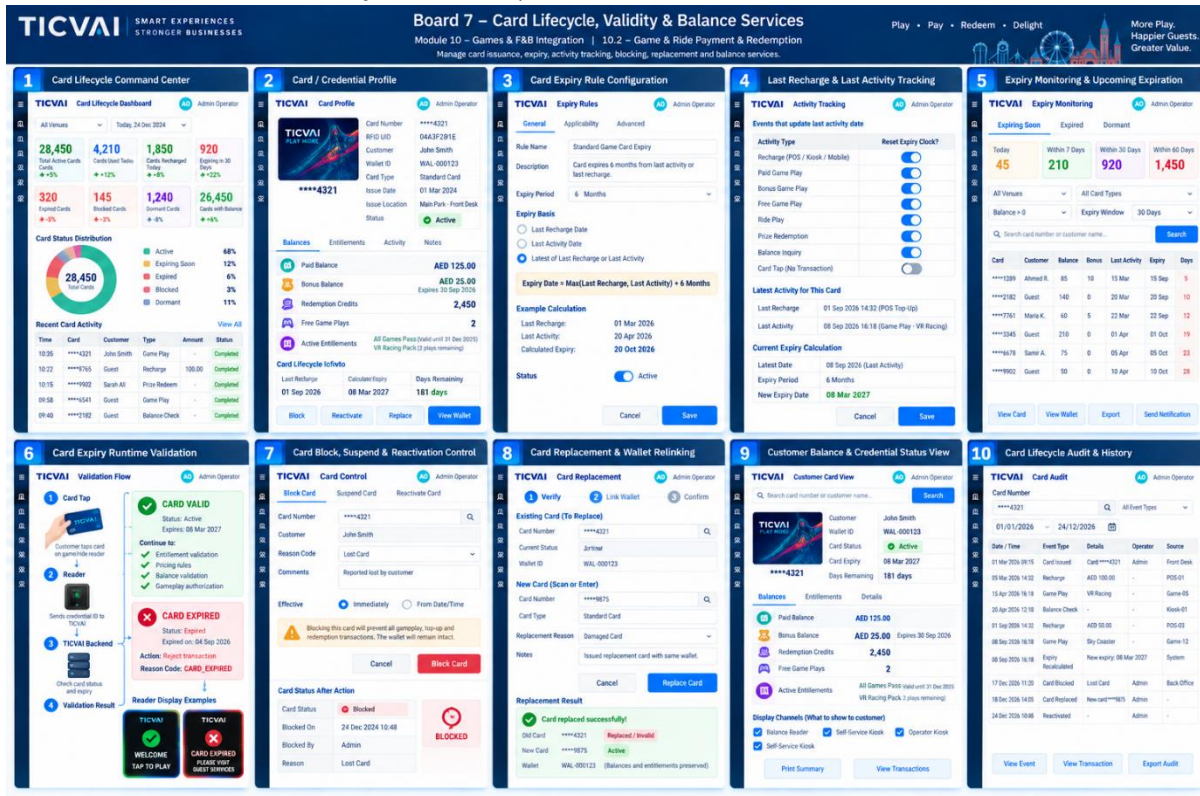
versus

Gameplay → Game Result → TICVAI → Wallet

Board 7 — Card Lifecycle, Validity & Balance Services

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption



Board Purpose: Manage the lifecycle of the customer game card / RFID credential used across games and rides, including card issuance status, wallet linkage, last recharge and activity tracking, automatic expiry calculation, balance visibility, blocking/reactivation, and card-level audit.

This board is primarily driven by requirement **10.2.21**, which states that a card should expire after **6 months based on the last card activity date or last recharge**. The module also requires a balance-check reader for customers.

Because the physical reader is only an edge device, **TICVAI owns the card lifecycle and expiry logic in the backend**. The reader simply reads the credential and receives the valid/expired/blocked decision from TICVAI.

The visual Board 7 must contain **exactly these 10 screens**.

Screen 1 — Card Lifecycle Command Center

Purpose

Give operations and management a central view of all active game cards / RFID credentials and their lifecycle status.

KPI Cards

- Total Active Cards
- Cards Used Today
- Cards Recharged Today
- Expiring in 30 Days
- Expired Cards
- Blocked Cards
- Dormant Cards
- Cards with Wallet Balance

Lifecycle Distribution

Show:

[Active](#) | [Expiring Soon](#) | [Expired](#) | [Blocked](#) | [Dormant](#)

Card Activity Table

Card	Customer	Balance	Last Recharge	Last Activity	Expiry	Status
****4321	John Smith	AED 125	01 Sep	08 Sep	08 Mar	Active
****6215	Guest	AED 35	01 Mar	04 Mar	04 Sep	Expired

Filters

- Venue
- Status
- Expiry Period
- Last Activity
- Last Recharge
- Customer/Card

Actions

[Search Card](#) | [View Card](#) | [Block](#) | [Reactivate](#) | [View Wallet](#)

Acceptance Criteria

Authorized users can identify cards approaching expiry, expired cards, blocked cards and currently active cards from a single screen.

Screen 2 — Card / Credential Profile

Purpose

Provide the master backend profile for an individual physical or digital game credential.

Card Information

- Card ID
- RFID UID / Credential ID
- Card Number
- Customer
- Wallet ID
- Issue Date
- Issue Location
- Card Type
- Status

Current Lifecycle Information

- Last Recharge
- Last Activity
- Calculated Expiry
- Days Remaining

Linked Values

- Paid Balance
- Bonus Balance

- Redemption Credits
- Free Plays
- Active Entitlements

Status Options

- Active
- Blocked
- Expired
- Suspended
- Replaced

Actions

Block | Reactivate | Replace | View Wallet | View Transactions

Acceptance Criteria

Operators can view the complete lifecycle and wallet association of a credential without storing business logic inside the physical card.

Screen 3 — Card Expiry Rule Configuration

Purpose

Configure the backend rule that determines when a card expires.

The source explicitly requires:

Card expiry of **6 months based on last card activity or last recharge**.

Rule Configuration

Rule Name: Standard Game Card Expiry

Expiry Period:

6 Months

Expiry Basis

Configure:

- Last Recharge Date
- Last Activity Date
- Latest of Last Recharge or Last Activity

Recommended interpretation

For the requirement as written, TICVAI should calculate the expiry from the **most recent qualifying event**:

Expiry Date = Max(Last Recharge, Last Activity) + 6 Months

Example

Last Recharge: **01 March 2026**

Last Activity: **20 April 2026**

Calculated Expiry:

20 October 2026

Applicability

- Venue
- Card Type
- Wallet Type

- Product
- Customer Segment

Acceptance Criteria

TICVAI automatically calculates a card's expiry date based on the configured inactivity/recharge rule.

Screen 4 — Last Recharge & Last Activity Tracking

Purpose

Maintain the two dates required for lifecycle calculation and show which events update them.

Last Recharge Events

Examples:

- POS Top-Up
- Kiosk Top-Up
- Mobile Top-Up
- Operator Top-Up

Last Activity Events

Potential qualifying events:

- Game Play
- Ride Play
- Wallet Transaction
- Redemption Transaction
- Balance Check, if configured to count
- Card/Reader Interaction, if configured

Configuration

For every activity type:

Activity	Reset Expiry Clock?
Recharge	✓
Paid Game Play	✓
Bonus Game Play	✓
Free Game Play	✓
Prize Redemption	✓
Balance Inquiry	Configurable

Current Calculation

Last Recharge: 01 Sep 2026

Last Activity: 08 Sep 2026

Expiry Basis: Last Activity

New Expiry: 08 Mar 2027

Acceptance Criteria

The system records qualifying recharge/activity events and recalculates expiry automatically.

TICVAI Recommended Enhancement: configurable qualifying-event list. The source defines expiry by activity/recharge but does not specify which activity types reset the period.

Screen 5 — Expiry Monitoring & Upcoming Expiration

Purpose

Identify cards approaching expiry so operators can monitor liability and customer impact.

Expiry Buckets

- Expiring Today
- Within 7 Days
- Within 30 Days
- Within 60 Days
- Expired

Table

Card	Balance	Bonus	Last Activity	Expiry	Days
****1289	85	10	15 Mar	15 Sep	5
****2182	140	0	20 Mar	20 Sep	10

Filters

- Balance > 0
- Bonus > 0
- Redemption Credits > 0
- Venue
- Card Type
- Expiry Window

Actions

[View Card](#) | [View Wallet](#) | [Export](#)

Acceptance Criteria

The operation can identify cards approaching expiration and see the value associated with those credentials.

TICVAI Recommended Enhancement: customer communication triggers can later be integrated with CRM/marketing without duplicating CRM configuration here.

Screen 6 — Card Expiry Runtime Validation

Purpose

Show how TICVAI automatically handles a card tap when the credential is active or expired.

Backend Validation Flow

Reader Tap

↓

Reader sends Credential ID

↓

TICVAI retrieves Card Profile

↓

Check Status

↓

Check Calculated Expiry

↓

Active Result**CARD VALID**

Expires: 08 Mar 2027

Continue to:

- Entitlement validation
- Pricing
- Balance validation
- Gameplay authorization

Expired Result**CARD EXPIRED**

Gameplay authorization stops.

Reader Response

TICVAI sends:

REJECT

Reason Code:

CARD_EXPIRED

Reader displays:

CARD EXPIRED**Acceptance Criteria**

Expired cards cannot initiate gameplay, while valid cards proceed automatically to the remaining transaction rules.

Screen 7 — Card Block, Suspend & Reactivation Control**Purpose**

Allow authorized operators to disable a lost, suspicious or invalid card without affecting the underlying wallet record.

Actions

- Block
- Suspend
- Reactivate
- Replace Card

Block Form

- Card
- Reason Code
- Comment
- Effective Immediately
- Operator

Reason Examples

- Lost Card
- Stolen Card
- Customer Request
- Fraud/Suspicious Activity
- Operational Issue

Backend Behavior

Blocked Card Tap

- Reader sends credential
- TICVAI recognizes blocked status
- Transaction rejected
- Wallet remains protected

Reactivation

Require:

- Permission
- Reason
- Audit entry

Acceptance Criteria

A card can be blocked or reactivated independently from its underlying wallet balance.

TICVAI Recommended Enhancement: lifecycle controls beyond expiry are needed operationally but are not explicitly detailed in 10.2.21.

Screen 8 — Card Replacement & Wallet Relinking

Purpose

Allow a damaged, lost or replaced physical card to be substituted while retaining the customer's wallet and entitlements.

Replacement Flow

Old Card

****4321

↓

Block Old Credential

↓

Issue / Scan New Credential

****9875

↓

Link Existing Wallet

↓

Preserve

- Paid Balance
- Bonus Balance
- Redemption Credits

- Entitlements
- Transaction History

Fields

- Existing Card
- Replacement Card
- Wallet
- Replacement Reason
- Operator
- Date/Time

Result

Old Credential:

REPLACED / INVALID

New Credential:

ACTIVE

Acceptance Criteria

Replacing the physical card does not require creating a new wallet or losing customer balances and entitlements.

TICVAI Recommended Enhancement: this is operationally necessary for a card-based amusement system but is not explicitly stated in the source matrix.

Screen 9 — Customer Balance & Credential Status View

Purpose

Provide the backend configuration/view that supports balance-check readers, operator kiosks and self-service kiosks.

The source requires a dedicated reader for checking customer balance and also requires wallet credits to be viewable at operator and self-service kiosks.

Customer View

Card: ****4321

Display

Paid Balance

AED 125

Bonus Balance

AED 25

Redemption Credits

2,450

Free Games

2

Entitlements

- All Games Pass
- VR Racing × 2

Card Information

Status: Active

Card Expiry: 08 Mar 2027

Bonus Expiry: 30 Sep 2026

Display Channels

Configure what is visible on:

- Balance Reader
- Self-Service Kiosk
- Operator Kiosk

Acceptance Criteria

The same TICVAI backend balance data can be securely presented through different customer-facing devices.

This does **not duplicate Board 2 Screen 10**. Board 2 configures the **physical balance reader/device**; this screen defines the **card/wallet information service and backend data presented through that reader**.

Screen 10 — Card Lifecycle Audit & History

Purpose

Maintain full traceability of card lifecycle changes.

Lifecycle Timeline

Example:

01 Mar 2026

Card Issued

05 Mar 2026

AED 100 Recharge

15 Apr 2026

Game Played

15 Oct 2026

Expiry Recalculated

17 Apr 2026

Card Blocked

18 Apr 2026

Replacement Card Assigned

Audit Fields

- Date/Time
- Card
- Event Type
- Previous Status
- New Status
- Wallet
- Operator/System
- Reason
- Source Device
- Transaction Reference

Events

- Card Issued
- Recharge
- Activity
- Expiry Recalculated
- Expired
- Blocked

- Reactivated
- Replaced
- Wallet Relinked

Actions

[View Event](#) | [View Transaction](#) | [Export Audit](#)

Acceptance Criteria

Every card lifecycle change can be traced to its originating event, device or authorized operator.

Board 7 — Exact Screen Mapping

The visual design for Board 7 must contain **exactly these ten screens and in this order**:

#	Screen
1	Card Lifecycle Command Center
2	Card / Credential Profile
3	Card Expiry Rule Configuration
4	Last Recharge & Last Activity Tracking
5	Expiry Monitoring & Upcoming Expiration
6	Card Expiry Runtime Validation
7	Card Block, Suspend & Reactivation Control
8	Card Replacement & Wallet Relinking
9	Customer Balance & Credential Status View
10	Card Lifecycle Audit & History

Important architecture for the design

The most important screen is **Screen 3**, because we need development team to understand that the **6-month expiry is calculated by TICVAI**, not programmed manually into each Chinese reader.

The architecture should appear clearly in **Screen 6**:

Card Tap → Chinese Reader → Credential ID → TICVAI Backend → Card Status + Expiry Validation → APPROVE/REJECT → Reader Display

Also, **Screen 9 must not look like another reader-configuration screen**. Board 2 already configures the physical reader. Board 7 configures and exposes the **backend balance/card-status information that the reader or kiosk requests**.

This makes Board 7 the backend answer to:

“Is this card still valid, what wallet does it belong to, what value does it contain, and what should happen when the card becomes inactive, expired, lost or replaced?”

Board 8 — Operations Monitoring, Transactions & Exception Control

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption



Board Purpose: Provide TICVAI operations teams with centralized real-time monitoring of the complete game/ride transaction ecosystem after configuration is deployed: reader communication, customer taps, gameplay authorization, wallet deductions, entitlement consumption, redemption activity, failed transactions, device issues, and operational exceptions.

This is the **operational monitoring board**, not another configuration board. Boards 1–7 define attractions, readers, wallet rules, entitlements, pricing, redemption, and card lifecycle. Board 8 answers:

“What is happening across the games and rides right now, and where is there a problem?”

This is important because the source requires real-time validation and balance deduction before gameplay, reader interaction, redemption transactions, and multiple operational rules that ultimately need traceability.

The visual Board 8 must contain **exactly these 10 screens**.

Screen 1 — Game & Ride Operations Control Center

Purpose

Provide the main real-time operational dashboard for the entire game/ride ecosystem.

KPI Cards

Today

- Total Customer Taps
- Authorized Plays
- Rejected Plays
- Wallet Value Consumed

- Bonus Value Consumed
- Entitlement Plays
- Redemption Credits Earned
- Reader/Device Issues

Live Operational Status

Show status by:

- Venue
- Zone
- Attraction
- Reader

Example:

Attraction	Reader	Status	Last Tap	Plays Today	Issues
VR Racing	R-014	Online	10:42	842	0
Basketball Pro	R-023	Online	10:42	612	2
Sky Coaster	R-001	Online	10:41	1,025	0
Prize Crane	R-031	Offline	10:25	321	1

Visual Zone Health

Main Park — Healthy

Arcade — Healthy

Fun Zone — Warning

Filters

- Client
- Venue
- Zone
- Attraction Type
- Attraction
- Reader
- Status

Acceptance Criteria

Operations can determine overall system health and transaction activity from one screen.

Screen 2 — Live Gameplay Transaction Monitor

Purpose

Display gameplay transactions as they occur across all connected readers.

Live Transaction Feed

Time	Card	Reader	Attraction	Price	Source	Result
10:42:18	****4321	R-014	VR Racing	20	Bonus + Wallet	Approved
10:42:16	****7621	R-001	Sky Coaster	—	All Games Pass	Approved
10:42:12	****9912	R-023	Basketball	20	Wallet	Rejected

Transaction Sources

- Paid Wallet
- Bonus
- Free Game
- Package
- Entitlement
- VIP
- Retry

Live Status

 **Approved**

 **Rejected**

 **Processing**

Transaction Detail Drawer

Selecting a transaction should show:

- Transaction ID
- Card/credential
- Wallet
- Reader
- Attraction
- Price rule
- Funding source
- Before balance
- Deduction
- After balance
- Authorization result
- Response time

Acceptance Criteria

Authorized operators can monitor gameplay transactions in near real time and open an individual transaction for investigation.

Screen 3 — Reader & Device Health Monitor

Purpose

Monitor communication and operational status of the physical Chinese readers and other connected devices.

Remember the architecture:

TICVAI Backend ↔ Reader ↔ Game/Ride

This screen monitors that connection; it does not configure business rules.

Device Status

Reader	Attraction	Connection	Last Heartbeat	Last Tap	Status
R-014	VR Racing	Online	5 sec ago	2 sec ago	Healthy
R-023	Basketball	Online	4 sec ago	8 sec ago	Healthy
R-031	Prize Crane	Offline	8 min ago	9 min ago	Critical

Health Indicators

- Online
- Offline
- Degraded
- Synchronizing
- Configuration Out-of-Sync
- Device Error

Device Details

- Reader ID
- Serial Number
- IP
- Firmware
- Configuration Version
- Last Sync
- Last Heartbeat
- Response Time

Actions

[View Device](#) | [Test Connection](#) | [Resync Configuration](#) | [View Logs](#)

Acceptance Criteria

Operations can immediately identify reader/device communication failures and configuration synchronization problems.

TICVAI Recommended Enhancement: centralized reader-health monitoring.

Screen 4 — Tap Validation & Decision Trace

Purpose

Allow operations to understand exactly how TICVAI reached an approval or rejection decision for an individual customer tap.

This relates directly to the requirement that balance, bonus and entitlements be validated in real time before gameplay.

Example Transaction

Transaction: TX-982411

Card: ****4321

Reader: R-014

Attraction: VR Racing

Decision Trace

1. Reader

✓ Valid

2. Card

✓ Active

3. Card Expiry

✓ Valid

4. Attraction

✓ Active

5. Entitlement

✗ Not Applicable

6. Bonus

✓ AED 10 Available

7. Paid Wallet

✓ AED 75 Available

8. Pricing

✓ AED 20

9. Deduction

AED 10 Bonus + AED 10 Wallet

10. Decision

✓ AUTHORIZED

Runtime Result**Reader Response:** APPROVE**Game Trigger:** SENT**Acceptance Criteria**

Operators can trace every major rule evaluated during an authorization decision.

Screen 5 — Rejected Transaction & Reason Analysis**Purpose**

Centralize all rejected gameplay transactions and identify why customers are being denied.

Rejection KPIs

- Rejections Today
- Rejection Rate
- Insufficient Balance
- Invalid/Expired Card
- Entitlement Failure
- Retap Protection
- Reader/Technical Failure

Rejection Reasons

Examples:

- INSUFFICIENT_BALANCE
- CARD_EXPIRED
- CARD_BLOCKED
- ENTITLEMENT_EXPIRED
- GAME_NOT_INCLUDED
- USAGE_LIMIT_REACHED
- RETAP_TOO_SOON
- READER_OFFLINE
- PRICE_NOT_FOUND

Rejection Table

Time	Attraction	Card	Reason	Reader
10:42	VR Racing	****3321	Insufficient Balance	R-014
10:40	Bumper Cars	****2178	Card Expired	R-007
10:38	Basketball	****6621	Retap Too Soon	R-023

Analysis

Show:

- Rejections by attraction
- Rejections by reader
- Rejections by reason
- Rejection trend

Acceptance Criteria

Operations can identify whether rejection spikes originate from customer conditions, configuration errors or technical problems.

Screen 6 — Wallet & Deduction Transaction Monitor

Purpose

Monitor the financial/value movement generated by game and ride activity.

The source requires the customer's amount to be deducted after the card tap based on available balance and the remaining balance to be updated.

Transaction Table

Transaction	Attraction	Price	Bonus	Paid	Remaining
TX-1001	VR Racing	20	10	10	65
TX-1002	Sky Coaster	35	0	35	120
TX-1003	Basketball	20	20	0	80

Summary

- Paid Credits Consumed
- Bonus Consumed
- Free Plays
- Entitlement Plays
- Retry Transactions
- VIP Transactions

Transaction Details

- Wallet ID
- Before Balance
- Deduction Source
- Amount
- After Balance
- Pricing Rule
- Reader
- Attraction
- Date/Time

Acceptance Criteria

Every gameplay authorization involving wallet value can be traced to its corresponding deduction.

Screen 7 — Entitlement & Free-Play Consumption Monitor

Purpose

Monitor consumption of packages, passes, limited plays, unlimited entitlements and free-game credits.

KPI Cards

- Entitlement Plays Today
- Free Plays
- Package Plays
- Unlimited Pass Uses
- Limited Plays Remaining
- Failed Entitlement Attempts

Consumption Table

Card	Product	Attraction	Before	Used	Remaining
****4321	Arcade Pack	VR Racing	2	1	1
****7122	Kids Pass	Carousel	Unlimited	1	Unlimited
****9102	Free Game	Basketball	1	1	0

Product Types

- All Games Pass
- Unlimited Specific Games
- Limited Specific Games
- Package
- Free Game Credit

The source explicitly distinguishes all-games access, unlimited selected-game access and limited selected-game access.

Acceptance Criteria

Operations can trace entitlement consumption and remaining usage without changing the entitlement configuration.

Screen 8 — Offline, Synchronization & Recovery Monitor

Purpose

Monitor readers or edge components that temporarily lose connectivity with TICVAI and manage recovery/synchronization.

This is a **TICVAI Recommended Enhancement**, based on the architecture we agreed for physical readers.

Status

- Online
- Offline
- Offline Transactions Pending
- Synchronizing
- Synchronized
- Conflict
- Failed Sync

Device Table

Reader	Offline Since	Pending TX	Config Version	Status
R-031	10:22	12	V2.4	Synchronizing
R-042	10:35	4	V2.4	Offline

Recovery Flow

Reader Offline

↓

Operate according to permitted cached rules

↓

Store transaction

↓

Connectivity restored

↓

Upload transactions

↓

TICVAI reconciliation

↓

Synchronization complete

Controls

- Offline Operation Allowed
- Maximum Offline Duration
- Maximum Offline Transaction Value
- Force Synchronization
- Block Offline Operation

Acceptance Criteria

Where supported by the selected reader/gateway architecture, TICVAI can identify offline devices and reconcile transactions after connectivity returns.

The exact offline capability must ultimately depend on the Chinese reader's SDK, local storage and communication architecture.

Screen 9 — Operational Alerts & Exception Center

Purpose

Provide one central queue for operational problems requiring attention.

Alert Categories

Reader

- Reader Offline
- Reader Not Responding
- Configuration Out-of-Sync

Transaction

- High Rejection Rate
- Duplicate Transaction
- Abnormal Retap Pattern

Wallet

- Deduction Failure
- Balance Mismatch

Pricing

- Missing Price
- Pricing Conflict

Redemption

- Redemption Credit Posting Failure
- Inventory Mismatch

Card

- Expiry/Status Issue

Alert Table

Severity	Alert	Attraction	Age	Status
Critical	Reader Offline	Prize Crane	12 min	Open
High	Rejection Spike	VR Racing	8 min	Investigating
Medium	Config Out-of-Sync	Basketball	4 min	Open

Workflow

Open → Assigned → Investigating → Resolved

Actions

Assign | Investigate | View Transaction | View Reader | Resolve

Acceptance Criteria

Operational exceptions can be prioritized, assigned and resolved through a central operational queue.

TICVAI Recommended Enhancement: centralized exception workflow.

Screen 10 — Operational Analytics & Reconciliation Dashboard

Purpose

Provide management and operations with consolidated performance analytics across gameplay, readers, wallets and redemption.

KPI Areas

Gameplay

- Total Plays
- Unique Players
- Authorization Rate
- Rejection Rate

Revenue / Wallet

- Paid Credit Consumed
- Bonus Consumed
- Average Spend Per Play

Entitlements

- Package Plays
- Free Plays
- VIP Plays
- Retry Plays

Devices

- Reader Uptime
- Offline Readers
- Average Response Time

Redemption

- Credits Earned
- Credits Redeemed
- Outstanding Redemption Credits

Analytics

- Plays by Attraction
- Revenue by Game/Ride
- Peak Usage Times
- Reader Performance
- Rejection Trend
- Wallet Consumption Mix
- Redemption Earned vs Redeemed

Reconciliation Summary

Gameplay Authorizations

vs.

Wallet Deductions

vs.

Reader/Game Starts

Example:

10,240 Authorized Plays
 10,238 Wallet/Entitlement Transactions
 10,238 Confirmed Game Starts

Variance: 2

Drill-Down

Click variance to investigate:

- Missing deduction
- Missing game-start confirmation
- Duplicate event
- Reader communication issue

Acceptance Criteria

Management can evaluate the operational and financial performance of the game/ride ecosystem and identify reconciliation discrepancies.

TICVAI Recommended Enhancement: automated transaction reconciliation and anomaly detection.

Board 8 — Exact Screen Mapping

The Board 8 visual must contain **exactly these 10 screens in this order**:

#	Screen
1	Game & Ride Operations Control Center
2	Live Gameplay Transaction Monitor
3	Reader & Device Health Monitor
4	Tap Validation & Decision Trace
5	Rejected Transaction & Reason Analysis
6	Wallet & Deduction Transaction Monitor
7	Entitlement & Free-Play Consumption Monitor
8	Offline, Synchronization & Recovery Monitor
9	Operational Alerts & Exception Center
10	Operational Analytics & Reconciliation Dashboard

Board 8 should **not introduce new business configuration**.

For example:

Board 2 configures the reader.

Board 3 configures wallet/bonus.

Board 4 configures validation and entitlements.

Board 5 configures pricing.

Board 6 configures redemption.

Board 7 configures card lifecycle.

Board 8 monitors all of them operating together.

The main runtime architecture that the design should communicate is:

Customer Card/RFID → Chinese Reader → TICVAI Backend → Card Validation → Entitlement → Pricing → Bonus/Wallet → APPROVE/REJECT → Reader → Game Controller → Transaction Confirmation → TICVAI Monitoring

This is the board that gives the operator the **real-time control-room view of the entire TICVAI game-reader ecosystem**.

Based on the original matrix, **Boards 1–8 already cover the functional requirements in 10.2.1–10.2.22**. So Board 9 should not invent another core transaction function. I recommend using it for a critical layer that becomes necessary once TICVAI is controlling third-party Chinese readers: **Reader Integration, Protocol & Edge Management**.

This is a **TICVAI recommended architecture board**, rather than a direct standalone requirement from the matrix. It turns the hardware/software architecture we clarified into something development team can actually build.

Board 9 — Reader Integration, Protocol & Edge Management

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption

Board Purpose: Provide the technical backend layer through which TICVAI integrates with different game/ride readers, manages communication protocols, maps reader commands and events, distributes configuration, receives transaction events, triggers games, manages connectivity, and supports future reader manufacturers without embedding TICVAI's business logic inside the hardware.

Core Architecture

TICVAI Business Rules

↓

TICVAI Reader Integration / Edge Layer

↓

Reader API / SDK / Protocol

↓

Chinese Reader

↓

Game / Ride Controller

The reader remains an execution device. **Pricing, wallet, entitlement, bonus, validity and other business decisions remain owned by TICVAI.**

Screen 1 — Reader Integration Command Center

Purpose

Provide the technical team with a centralized view of every reader integration deployed across TICVAI clients and venues.

KPI Cards

- Total Integrated Readers
- Online
- Offline
- Integration Errors
- Pending Configuration Sync
- Transactions Today
- Average Response Time
- Protocol Errors

Integration Overview

Reader	Model	Protocol	Attraction	Status	Config
R-001	GR-X10	TCP/IP	Sky Coaster	Online	Synced
R-014	GR-X10	TCP/IP	VR Racing	Online	Synced
R-023	GR-X20	API	Basketball	Online	Pending
R-031	GR-X20	API	Prize Crane	Offline	Out-of-Sync

Filters

- Client
- Venue
- Manufacturer
- Model
- Protocol
- Status
- Configuration Version

Actions

[Add Integration](#) | [Test Connection](#) | [Synchronize](#) | [View Logs](#) | [Open Reader](#)

Acceptance Criteria

Technical administrators can see the health and synchronization state of all reader integrations centrally.

Screen 2 — Reader Manufacturer & Model Profile

Purpose

Allow TICVAI to support multiple reader manufacturers/models without building business rules specifically for one Chinese reader.

Manufacturer Profile

- Manufacturer
- Model
- Device Family
- Firmware
- SDK Version
- API Version
- Supported Protocols
- Certification Status

Hardware Capabilities

Configure supported capabilities:

- ☒ RFID
- ☒ NFC
- ☒ Screen
- ☒ LED / RGB Lighting
- ☒ Sound
- ☒ Touch Screen
- ☒ Local Storage
- ☒ I/O Output
- ☒ Network Connectivity

Runtime Capabilities

- ☒ Display Price
- ☒ Display Wallet Balance
- ☒ Display Messages
- ☒ Change LED Color
- ☒ Trigger Game
- ☒ Receive Game Result
- ☒ Cache Configuration
- ☒ Store Offline Transactions

Acceptance Criteria

TICVAI can maintain capability profiles for different reader models and know which functions each device supports.

Architectural benefit

This prevents TICVAI from becoming dependent on **one Chinese hardware supplier**.

Screen 3 — Communication Protocol Configuration

Purpose

Configure how TICVAI communicates with each reader/device family.

Supported Integration Methods

Depending on hardware capabilities:

- TCP/IP
- REST API
- WebSocket
- MQTT
- SDK
- Serial / RS-232/485
- Other vendor protocol

Connection Configuration

- Protocol
- Host / IP
- Port
- Endpoint
- Authentication
- Timeout
- Retry Count
- Heartbeat Interval
- Encryption
- Connection Mode

Example

Protocol: TCP/IP

Host: 192.168.10.55

Port: 9100

Timeout: 3 seconds

Heartbeat: 10 seconds

Connection Test

TEST CONNECTION

Result:

 Connected

Response: 42 ms

Acceptance Criteria

Technical administrators can configure and validate communication between TICVAI and supported reader hardware.

Screen 4 — Reader Command & Event Mapping

Purpose

Translate TICVAI's standardized commands into the specific commands understood by each reader manufacturer.

This is one of the most important architectural screens.

TICVAI should have standardized commands such as:

- DISPLAY_PRICE
- DISPLAY_MESSAGE
- LED_COLOR
- APPROVE_PLAY
- REJECT_PLAY
- TRIGGER_GAME
- SHOW_BALANCE
- SYNC_CONFIG

Reader Events

Readers can return standardized events such as:

- CARD_TAPPED
- GAME_STARTED
- GAME_COMPLETED
- SCORE_RECEIVED
- PRIZE_WON
- DEVICE_ERROR
- HEARTBEAT

Mapping Example

TICVAI Command

APPROVE_PLAY

↓

Vendor GR-X10 Command

CMD:PLAY:01

Acceptance Criteria

TICVAI uses a normalized internal command/event model while vendor-specific adapters translate it to/from the reader's protocol.

Why this matters

If you replace the Chinese reader later, **Boards 1–8 do not need to change**. Only the hardware adapter changes.

Screen 5 — Reader Configuration Deployment & Synchronization

Purpose

Push backend configuration from TICVAI to readers where local configuration is required.

Configuration Package

Potentially include:

- Reader ID
- Attraction
- Display Theme

- Display Messages
- LED Rules
- Price Display
- Retap Settings
- Device Parameters
- Offline Parameters
- Configuration Version

Deployment

TICVAI Configuration V2.8

↓

Readers selected: **126**

↓

Deploy

Status

Reader	Current	Target	Status
R-001	V2.8	V2.8	Synced
R-014	V2.8	V2.8	Synced
R-023	V2.7	V2.8	Pending
R-031	V2.6	V2.8	Failed

Actions

Deploy | Retry | Rollback | Compare Versions

Acceptance Criteria

TICVAI can identify which configuration version is running on each reader and synchronize supported settings.

Screen 6 — Game Trigger & I/O Control Mapping

Purpose

Configure how an approved TICVAI transaction physically starts a game or ride.

Flow

Card Tap

- TICVAI Authorization
- **APPROVED**
- Reader receives command
- Reader sends output
- Game Controller
- **GAME START**

Output Configuration

Depending on hardware:

- Relay Output

- Digital Output
- Pulse
- TCP/IP Command
- Serial Command
- API Command

Example

Basketball Pro

Trigger Type: Relay Pulse

Output: DO-1

Pulse Duration: 500 ms

Feedback Input

Potential inputs:

- Game Started
- Game Completed
- Prize Won
- Fault
- Score Result

Acceptance Criteria

TICVAI authorization can be mapped through the reader to the appropriate physical game-start mechanism.

Screen 7 — Reader Screen, LED & Sound Output Mapping

Purpose

Map TICVAI transaction results to the physical reader's screen, lighting and sound capabilities.

This supports the matrix requirement for free-game color indication and custom reader themes.

Result Mapping

AUTHORIZED

- Screen: PLAY AUTHORIZED
- LED: Green
- Sound: Success Tone

FREE PLAY

- Screen: FREE PLAY — TAP TO START
- LED: Configured Free-Play Color
- Sound: Free Play Tone

VIP

- Screen: VIP PRICE AED 15
- LED: VIP Theme

REJECTED

- Screen: Reason Message
- LED: Red
- Sound: Error Tone

Preview

The screen should show a physical reader mockup with:

NORMAL / FREE / VIP / REJECTED

states.

Acceptance Criteria

TICVAI can translate backend transaction results into supported reader display, LED and sound behavior.

Screen 8 — Edge Cache & Offline Rule Package

Purpose

Configure what information can be securely distributed to the edge/reader or local gateway for temporary offline operation.

Offline Package

Depending on hardware architecture:

- Reader Identity
- Attraction ID
- Price Rules
- Card/credential validation data
- Entitlement data
- Retap rules
- Offline limits
- Security keys
- Configuration version

Controls

Allow Offline Operation: ON

Maximum Offline Duration: 2 hours

Maximum Transaction Count: 500

Maximum Value Exposure: AED 5,000

Security

- Signed configuration
- Encryption
- Package expiry
- Device binding
- Version validation

Offline Flow

Cloud unavailable

- Edge detects outage
- Valid cached rules used
- Transactions stored locally
- Connectivity restored
- Transactions uploaded
- TICVAI reconciliation

Acceptance Criteria

Where supported by the selected reader/gateway, controlled offline operation can continue without allowing unrestricted financial exposure.

TICVAI Recommended Enhancement.

Screen 9 — Device Diagnostics & Integration Logs

Purpose

Give development team / support teams detailed technical visibility when a reader or game is not operating correctly.

Diagnostics

- Connection Status
- Last Heartbeat
- API/Protocol Status
- Response Time
- Configuration Version
- Firmware Version
- Last Successful Transaction
- Last Error

Event Log

Time	Direction	Event	Result
10:42:01	Reader → TICVAI	CARD_TAPPED	Received
10:42:01	TICVAI → Reader	APPROVE_PLAY	Sent
10:42:02	Reader → Game	TRIGGER_GAME	Success
10:42:03	Reader → TICVAI	GAME_STARTED	Received

Error Example

GAME_TRIGGER_FAILED

Reader received approval but no game-start confirmation was returned.

Actions

[Ping](#) | [Test Reader](#) | [Test Display](#) | [Test LED](#) | [Test Game Trigger](#) | [Download Logs](#)

Acceptance Criteria

Technical teams can trace communication from TICVAI through the reader to the game/controller and identify the point of failure.

Screen 10 — Integration Certification & Test Console

Purpose

Provide a controlled environment for testing and approving a new reader model before deployment.

This is particularly important for your plan because you will purchase the hardware separately and TICVAI must verify that the reader supports everything required by the platform.

Test Suite

Connectivity

- Connection
- Authentication
- Heartbeat

Card

- RFID Tap
- NFC Tap
- Credential Reading

Display

- Text
- Price
- Balance

LED

- Normal
- Free
- VIP
- Reject

Gameplay

- Authorization
- Game Trigger
- Game Started
- Game Completed

Transactions

- Paid Play
- Bonus
- Entitlement
- Retry
- VIP

Offline

- Cache
- Transaction Storage
- Re-Synchronization

Test Result**GR-X20 Reader**

Connectivity ✓
 RFID ✓
 Display ✓
 LED ✓
 Game Trigger ✓
 Offline ✓
 Reconciliation ✓

Certification Status**APPROVED FOR TICVAI****Actions**

Run Full Test | Run Selected Tests | Generate Test Report | Approve Reader

Acceptance Criteria

A reader model cannot be marked as TICVAI-certified until all mandatory integration tests have passed.

Board 9 — Exact Screen Mapping

#	Screen
1	Reader Integration Command Center
2	Reader Manufacturer & Model Profile
3	Communication Protocol Configuration
4	Reader Command & Event Mapping
5	Reader Configuration Deployment & Synchronization
6	Game Trigger & I/O Control Mapping
7	Reader Screen, LED & Sound Output Mapping
8	Edge Cache & Offline Rule Package
9	Device Diagnostics & Integration Logs
10	Integration Certification & Test Console

Why I recommend keeping Board 9

Unlike Boards 1–8, **Board 9 is not directly specified as a standalone function in the provided matrix**. It is the technical architecture required to implement the hardware-independent approach we discussed.

The key design principle is:

Never build TICVAI around the protocol of one reader manufacturer.

Instead:

TICVAI Core Business Engine

↓

Standard TICVAI Reader API / Device Abstraction Layer

↓

Vendor Adapter A / Vendor Adapter B / Vendor Adapter C

↓

Chinese Reader / Future Reader

This means that if your first Chinese reader turns out to be poor quality or the supplier changes its commercial terms, **you can replace the reader without redesigning your wallet, pricing, entitlement, redemption, or game-management platform**.

That separation could become one of the most important technical decisions in this entire module.

Board 10 — Customer Self-Service, Kiosk & Game Wallet Experience

Module: 10 – Games & F&B Integration

Sub-Module: 10.2 – Game & Ride Payment & Redemption



Board Purpose: Configure and manage the customer-facing self-service experience through kiosks, balance stations, and supported reader screens, allowing guests to identify their card/wallet, view balances and entitlements, top up value, understand bonus/free-play benefits, check redemption credits, and review recent game transactions without requiring an operator.

This board completes an area that is explicitly present in the matrix but has not yet been given its own UI experience layer: the source requires credits stored in a digital wallet to be visible through **operator kiosks and self-service kiosks**, in addition to a dedicated balance-check reader.

It is important that this board does **not** recreate wallet, pricing, entitlement, or bonus rules. Those remain in Boards 3–7. Board 10 determines **what the customer sees and can do through the self-service channel**.

The design must contain **exactly these 10 screens**.

Screen 1 — Self-Service Experience Command Center

Purpose

Provide administrators with a centralized view of all customer-facing game wallet kiosks, balance stations, and enabled self-service channels.

KPI Cards

- Total Self-Service Devices
- Online Devices
- Offline Devices
- Customer Sessions Today
- Balance Checks

- Top-Ups
- Wallets Viewed
- Failed Sessions

Channel Overview

Device	Type	Venue	Sessions	Last Activity	Status
KSK-01	Self-Service Kiosk	Main Park	842	10:42	Online
BAL-05	Balance Station	Arcade	1,240	10:42	Online
KSK-08	Self-Service Kiosk	Family Zone	611	10:40	Online

Filters

- Client
- Venue
- Zone
- Device Type
- Status

Actions

[Configure Experience](#) | [View Device](#) | [Preview Customer Flow](#) | [View Sessions](#)

Acceptance Criteria

Administrators can monitor the status and usage of all self-service game-wallet channels centrally.

Screen 2 — Self-Service Kiosk Profile & Channel Configuration

Purpose

Define the business role of each kiosk or customer-facing station.

Kiosk Profile

- Kiosk ID
- Name
- Venue
- Zone
- Device Type
- Language
- Currency
- Status

Available Functions

Toggle:

- ☒ Check Balance
- ☒ View Bonus
- ☒ View Redemption Credits
- ☒ View Free Games
- ☒ View Entitlements
- ☒ Top Up Wallet
- ☒ View Recent Transactions
- ☒ View Card Expiry

Optional:

- Replace/issue card workflow

- Customer support request

Access Method

- RFID Tap
- NFC Tap
- QR
- Card Number
- Digital Wallet Credential

Acceptance Criteria

Administrators can determine which self-service functions are available on each kiosk/device.

Screen 3 — Customer Card / Wallet Identification

Purpose

Configure the first customer step: identifying the customer's game card or digital wallet securely.

Welcome Screen

WELCOME

Tap your card to continue.

Identification Methods

- RFID
- NFC
- QR
- Card Number
- Digital Credential

Runtime Flow

Customer Tap

↓

Reader captures Credential ID

↓

TICVAI identifies Wallet

↓

Validate Card Status

↓

Open Customer Session

Invalid Results

- Card Not Found
- Card Blocked
- Card Expired
- Wallet Inactive

Example

Card recognized

****4321

Welcome, John

Acceptance Criteria

A self-service session opens only after TICVAI successfully identifies and validates the customer's credential.

Screen 4 — Customer Wallet & Balance Summary

Purpose

Present all relevant wallet balances clearly to the customer.

The source specifically requires customers' stored credits to be viewable through self-service kiosks.

Customer Balance Cards

Paid Credit

AED 125

Bonus Credit

AED 25

Expires: 30 Sep

Redemption Credits

2,450

Free Games

2

Active Entitlements

- All Games Pass
- Kids Unlimited Ride Pass
- Arcade Package — 2 Plays Remaining

Card Information

- Card Status
- Card Expiry
- Last Recharge

Actions

[Top Up](#) | [View Benefits](#) | [View Transactions](#) | [Finish](#)

Acceptance Criteria

Customers can distinguish paid credit, bonus, redemption credits, free games, and entitlements in a single view.

Screen 5 — Self-Service Wallet Top-Up

Purpose

Allow customers to add value to their game wallet through an enabled kiosk/channel.

Current Balance**AED 125****Top-Up Options**

AED 50

AED 100

AED 200

AED 500

or

Other Amount**Promotional Result**

Example:

TOP UP AED 100Paid Credit: **AED 100**Bonus: **AED 25**Total Added: **AED 125**

The actual top-up/bonus business rule comes from Board 3. The kiosk merely presents the applicable rule. The source gives the example of AED 100 top-up plus AED 25 bonus.

Payment

- Card Payment
- Supported Digital Payment
- Other configured payment method

Confirmation**TOP-UP SUCCESSFUL**

Paid Balance: AED 225

Bonus Balance: AED 50

Acceptance Criteria

The self-service channel retrieves the active TICVAI top-up rules and applies the resulting wallet and bonus values correctly.

Screen 6 — Bonus, Free Game & Benefit View**Purpose**

Help customers understand promotional value and benefits available in their wallet.

Bonus Section**Bonus Credit: AED 25**

Valid Until: 30 Sep 2026

Can be used at:

- ✓ Games
- ✓ Rides

Cannot be used at:

- ✗ F&B
- ✗ Retail

The source explicitly states that bonus may be restricted to games/rides and excluded from F&B and Retail.

Free Plays

Birthday Free Play

Remaining: 2

Available at:

- VR Racing
- Basketball Pro
- Bumper Cars

Entitlements

Kids Unlimited Ride Pass

Valid Today

Unlimited Usage

Acceptance Criteria

Customers can understand what promotional/free value they hold, where it can be used, and its remaining validity.

Screen 7 — Game & Ride Eligibility / “What Can I Play?”

Purpose

Allow guests to see which games and rides they can currently access based on their wallet, packages, free plays and entitlements.

Customer Position

Wallet: AED 125

Bonus: AED 25

Free Plays: 2

Package: Arcade Pass

Attraction Results

Attraction	Normal Price	Your Access	Status
VR Racing	AED 20	Included ×2	Play
Sky Coaster	AED 35	AED 35	Play
Basketball	AED 20	Free Play	Play
Premium Crane	AED 15	AED 15	Play

Filters

- Rides
- Video Games
- Skill Games
- Free for Me
- Included in My Package

Status

- Included
- Free
- Pay to Play
- Not Eligible

Acceptance Criteria

Customers can identify available gameplay options using the real-time rules configured in TICVAI.

TICVAI Recommended Enhancement: customer-facing eligibility discovery.

Screen 8 — Redemption Balance & Prize Discovery

Purpose

Allow customers to check redemption credits and browse prizes they can afford before visiting the redemption counter.

Redemption Balance

2,450 Credits

Prize Catalogue

Affordable Now

- Teddy Bear — 800
- Football — 1,000
- Water Bottle — 300

Locked / More Credits Needed

Headphones

3,500 Credits

You need 1,050 more credits.

Filters

- Available Now
- Prize Category
- Credit Range
- Venue

Stock Indicator

- Available
- Low Stock
- Out of Stock

Important

Actual prize redemption remains on Board 6. This screen is customer discovery/view only unless a future self-service redemption capability is approved.

Acceptance Criteria

Customers can see their redemption balance and identify prizes that are currently available within that balance.

Screen 9 — Customer Game & Wallet Transaction History

Purpose

Allow the customer to review recent activity associated with the card/wallet.

Recent Transactions

Time	Activity	Amount
10:42	VR Racing	-AED 20
10:30	Bonus Added	+AED 25
10:30	Top-Up	+AED 100
10:15	Basketball	Free Play
09:50	Prize Credits Earned	+200

Transaction Detail

- Attraction
- Reader
- Transaction Type
- Price
- Funding Source
- Credits Earned
- Remaining Balance
- Date/Time

Filters

- Games/Rides
- Top-Ups
- Bonus
- Redemption
- Free Play

Acceptance Criteria

Customers can review recent wallet/game activity without requiring back-office access.

Screen 10 — Self-Service UI Theme, Language & Journey Configuration

Purpose

Allow TICVAI administrators to configure the customer-facing kiosk experience without changing business logic.

Experience Configuration

Branding

- Client Logo
- Background
- Welcome Message
- Hero Image

Languages

- English
- Arabic
- Other configured languages

Home Menu

Allow administrators to determine the order of:

1. Check Balance

2. Top Up
3. My Benefits
4. What Can I Play?
5. Redemption
6. Transactions

Session Configuration

- Session Timeout
- Auto Logout
- Confirmation Timeout
- Privacy Reset After Session

Reader/Kiosk Display Preview

Desktop-style preview showing:

WELCOME TO THE PARK

Tap Your Card

Accessibility

- Large Text
- High Contrast Mode
- Audio Guidance, where hardware supports it

Publish

Draft → Preview → Publish

Apply to:

- Selected Kiosk
- Venue
- Kiosk Group
- All Client Kiosks

Acceptance Criteria

The customer-facing experience can be branded and configured centrally while continuing to consume the same TICVAI backend wallet and gameplay services.

TICVAI Recommended Enhancement: centralized journey/theme configuration.

Board 10 — Exact Screen Mapping

#	Screen
1	Self-Service Experience Command Center
2	Self-Service Kiosk Profile & Channel Configuration
3	Customer Card / Wallet Identification
4	Customer Wallet & Balance Summary
5	Self-Service Wallet Top-Up
6	Bonus, Free Game & Benefit View
7	Game & Ride Eligibility / “What Can I Play?”
8	Redemption Balance & Prize Discovery

#	Screen
9	Customer Game & Wallet Transaction History
10	Self-Service UI Theme, Language & Journey Configuration

Important scope boundary

Board 10 is the **experience layer**, not another business-rule engine.

For example:

Board 3 decides that AED 100 top-up gives AED 25 bonus.

Board 10 shows that offer to the customer.

Board 4 decides whether the customer's package allows VR Racing.

Board 10 displays “**Included — 2 Plays Remaining.**”

Board 6 owns redemption credits and prize inventory.

Board 10 allows the customer to see **2,450 credits** and which prizes they can afford.

Board 7 calculates whether the card expires in six months.

Board 10 simply displays the resulting expiry date.

Why Board 10 is useful

Unlike some of our architecture-driven boards, this one has a direct foundation in the original requirement because the matrix explicitly calls for wallet-credit visibility through **operator and self-service kiosks**.

However, functions such as “**What Can I Play?**”, **prize browsing, configurable customer journey, branding, and detailed self-service transaction history** are **TICVAI Recommended Enhancements** rather than explicit requirements in 10.2.

This gives us a clean separation:

TICVAI Backend Rules → TICVAI APIs → Self-Service Kiosk → Customer

The kiosk never owns the wallet or gameplay logic; it is simply another controlled interface consuming the same TICVAI backend services.